FGNS-TRM: A Hybrid Geometric-Analytical Prototype Exploring Structural Priors, Non-Abelian 3D Frames, Cellular Sheaves & Cauchy Recurrence on ARC-AGI-2

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Codebase & License: Open-Source Release (`fgns_engine` v5.0.0) under **MIT-0 (MIT No Attribution)**
Hardware Target: Single Consumer GPU (`NVIDIA GeForce RTX 5070` , 12 GB GDDR7, Blackwell `sm_120` ; ~8 GB peak training VRAM) and Kaggle `L4x4` (`sm_89`)

Abstract

Many competitive approaches to the Abstraction and Reasoning Corpus (`ARC-AGI-2`) employ Large Language Model (LLM) program synthesis, hand-crafted Domain-Specific Languages (DSLs), or hybrid neural-symbolic search. In this work, we explore a complementary geometric-analytical research direction within **Structural Computation**: *can spatial reasoning be expressed through compact, continuous-state geometric and topological operators—trained from scratch without LLMs or symbolic DSL search—that embed canvas boundaries, palette-quota constraints, disentangled 3D rotational symmetry, directional propagation, and cellular connectivity directly into their computational substrate?* As a first experimental realization of this program, we introduce the **Finsler-Gauge Cellular Sheaf Network (`FGNS-TRM v5.0.0`)**, a **6.57M-parameter (`6,574,411` exact trainable parameters)** hybrid prototype built around four core mechanisms:

1. **Two-Stage Upstream Structural Priors (`BoundaryMetricSizePredictor` & `EquivariantPaletteHistogramPredictor`):** A naive 30×30 padded grid representation with 11 cell states (colors $0 \dots 9$ plus exterior `PAD=10`) admits up to $11^{900} \approx 1.79 \times 10^{937}$ unconstrained configurations. Before pixel-level decoding, **Stage I** resolves target canvas dimensions $(H^\star, W^\star)$ over $\mathbb{Z}_{61} \times \mathbb{Z}_{61}$ via a Shared-SRAM CUDA flood-fill/ray-casting kernel, coupled Boltzmann-Gibbs hypothesis voting, and a Woodbury-Tikhonov 1-Jet Stokes resolvent (`Method 1`). **Stage II** conditions on the predicted area $K_{\text{cap}} = H^\star \times W^\star$, using S_{10}-equivariant 8D structural role descriptors, a diode annihilation sink for vanishing auxiliary colors, and straight-through Hamilton-Hare Largest Remainder Quota (`LRQ`) apportionment to estimate integer color quotas $\sum_{c=0}^{9} N_c^\star = K_{\text{cap}}$, which are injected into the decoder as a **soft Potts chemical potential bias**.
2. **Orthogonal Color-Geometry Subspace Splitting & Non-Abelian 3D Frames (`NA3D-Finsler-Lie`):** Two-pass Modified Gram-Schmidt (`MGS2`) constructs a numerically orthonormal palette basis in IEEE-754 FP32 ($\|QQ^\top - I_{11}\|_{\max} = 2.38 \times 10^{-7}$), splitting the 128D latent space into an 11D palette subspace and an orthogonal 117D geometric complement: $\mathbb{R}^{128} = \mathcal{H}^{(11)}_{\text{color}} \oplus \mathcal{H}^{(117)}_{\text{geom}}$ (`Lemma 1`). For projected spatial updates, this orthogonality reduces measured rotational color-subspace leakage to residual FP32 rounding error (1.05×10^{-7} vs. 0.7881 for unsplit SO(3) across our 25-task suite, and 1.54×10^{-8} vs. 0.2739 on unit-prototype patch tests, preserving 100.0% patch accuracy vs. 28.9% under unsplit rotation). Within $\mathcal{H}^{(117)}_{\text{geom}}$, each cell is parameterized as $C = 8$ Non-Abelian Asymmetric 3D Token (`NA3D`) simplices $(\mathbf{L}, \mathbf{C}, \mathbf{R}, \beta)$ evolving via SO(3) Rodrigues rotations and asymmetric Finsler-Randers proper time.
3. **8-Axis `OctoWave` , Stiefel Cellular Sheaves, Object Slots & Spectral-Green Operator:** Line-parallel 8-directional CUDA C++17 state-space scans (with discrete Householder specular wall reflection, achieving a post- `RMSNorm` diagonal impulse-response norm of 11.3104 at the opposite corner of an 8×8 probe vs. 0.0000 for a horizontal 2-axis scan) and edge-stopping Stiefel cellular sheaf diffusion on $\mathcal{V}_{32}(\mathbb{R}^{128})$ (`Lemma 2` , empirical Dirichlet energy ratio 0.707 vs. 7.840 blowup for sign-inverted anti-diffusion) couple with 12 Schur-complement object slots, a Leave-One-Out (`LOO`) Holographic Support Tape using SO(3) parallel transport, a non-linear J-involutive antisymmetric interaction operator (`True XOR`), and a 23-channel `DifferentiableSpectralGreenOperator` (`SpectralGreenWoodburyFunction` , verified to 7.76×10^{-10} max relative Jacobian error in `float64` `torch.autograd.gradcheck`).
4. **Dual-State CfC/Cauchy Recurrence & MDL-Inspired Schatten-1 `LOO-TTT` :** Separate thought and draft states $(Z^{(\tau)}, P^{(\tau)})$ evolve via proper-time dilated Cauchy resolvents across $N = 4$ steps (+5.66% pixel accuracy over $T = 1$), while task-specific operator weight perturbations $\Delta W = W - W_0$ adapt during Leave-One-Out test-time training (`Method 2`) under an MDL-inspired Newton-Schulz Schatten-1 nuclear norm penalty (compressing the Schatten-1/Frobenius participation effective rank r_{eff} from 3.46 to 1.10 and reducing 5-task continuous 8-step query loss from 0.4981 to 0.4736).

Evaluated on the **ARC-AGI-2 Public Evaluation set** (single-pass transductive inference with 0 gradient TTT steps after 1 epoch / 1,000 steps of pretraining on `arc-agi_training_challenges.json`), Stage I achieves **76.74% (132/172) exact dimension accuracy** across all 120 tasks (**88.0% with 0.40 px MAE on our 25-task ablation suite** vs. 0.0% for an unconstrained spatial MLP), Stage II cuts palette Earth Mover's Distance by **59.1%** ($1.3488 \to 0.5523$) and color hallucinations by **3.7×** ($1.26\% \to 0.34\%$), and the network solves compound dihedral occlusion (`0934a4d8`, $30 \times 30 \to 9 \times 3$) and topological cross filtering (`1818057f`, 22×22) at **100.0% exact match**, while reaching **98.93% (`832/841` px)** on Bravais lattice repair (`135a2760`) and **99.44% (`895/900` px)** on Eikonal flooding (`13e47133`, `100.0%` on the pure Eikonal branch) within **~7.8–8.2 GB peak training VRAM** on one RTX 5070 GPU. We conclude with a transparent diagnostic analysis of prototype limitations and a three-pillar research agenda for Structural Computation.

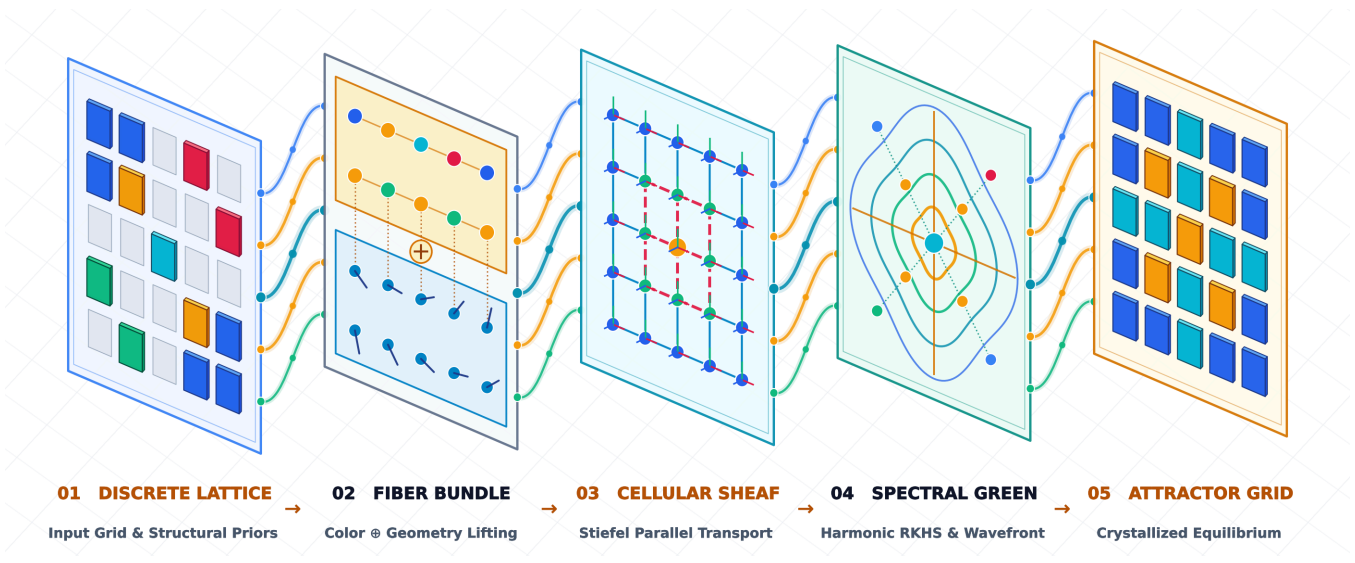


Figure 0 | Overview of the five-stage continuous geometric-analytical pipeline in `FGNS-TRM v5.0.0`: (01) Discrete Lattice & Two-Stage Upstream Structural Priors (`Stage I` boundary-metric size prior and `Stage II` equivariant palette quota prior), (02) Orthogonal Color-Geometry Fiber Bundle $\mathcal{H}_{\text{color}}^{(11)} \oplus \mathcal{H}_{\text{geom}}^{(117)}$, (03) Stiefel Cellular Sheaf Parallel Transport, (04) Unified Differentiable Spectral-Green RKHS & 8-Axis `OctoWave` Wavefront Operator, and (05) Dual-State Cauchy Crystallized Attractor Grid.

1. Introduction: From Pattern Memorization to Structural Computation

1.1. The Structural Bottleneck of Standard Deep Learning on `ARC-AGI-2`

In *On the Measure of Intelligence* (Chollet, 2019) and the *ARC Prize Technical Report* (Chollet et al., 2025/2026), Chollet et al. argue that genuine intelligence cannot be measured by skill performance on familiar training distributions alone; rather, it is the efficiency of skill acquisition on novel tasks grounded in **Core Knowledge priors**:

1. **Objectness & Cohesion:** Parsing discrete grids into persistent, bounded entities that move, occlude, or interact without dissolving into unstructured pixel clouds.
2. **Elementary Geometry & Symmetry:** Perceiving translations, rotations, reflections (the dihedral group D_4), aspect-ratio homotheties, and 2D periodic Bravais lattices.
3. **Topology & Connectivity:** Recognizing inside versus outside (Jordan curve separation), enclosed cavities, 1-pixel impermeable walls, and geodesic shortest paths around obstacles.
4. **Cardinal Arithmetic & Quota Conservation:** Counting discrete symbols, apportioning exact integer color budgets, and annihilating temporary marker masks.

When standard continuous neural backbones—such as Vision Transformers (Dosovitskiy et al., 2021; Hu et al., 2025), 2D Rotary Position Embedding (RoPE) Transformers (Su et al., 2024), or 2-axis State-Space Models like Mamba-2 (Dao & Gu, 2024)—are trained from scratch for direct grid-to-grid transduction without LLMs or DSL program search, they encounter three fundamental structural obstacles:

- **Obstacle I (The 11^{900} Unconstrained Canvas & Palette Space):** Embedding variable-size grids (1×1 to 30×30) into a padded 30×30 canvas over an 11-symbol vocabulary ($\{0, \dots, 9\} \cup \{\text{PAD} = 10\}$) spans $11^{900} \approx 1.79 \times 10^{937}$ states. Standard pixel-wise decoders attempt to infer the canvas boundary $\partial\Omega$, the global color cardinality histogram, and the interior spatial layout simultaneously within a single terminal projection.
- **Obstacle II (Entanglement of Color Identity and Spatial Geometry):** In a flat vector representation $\mathbf{h}(u) \in \mathbb{R}^{128}$, discrete color identity and continuous spatial geometry share the same coordinate axes. Even when

positional encodings such as 2D RoPE apply frequency-banded rotations within attention heads, subsequent dense learned weight matrices (W_{proj}, FFN layers, and test-time adaptation updates) mix all 128 coordinates across layers, causing spatial rotations or reflections to perturb inner products with discrete palette prototypes.

- **Obstacle III (Directional Blind Spots and Isotropic Boundary Bleeding):** Raster-order 2-axis state-space scans propagate information only along horizontal and vertical axes, exhibiting zero single-pass causal response along 45° diagonals and lacking specular reflection at obstacle walls. Conversely, standard isotropic graph convolutions (GCNs; Kipf & Welling, 2017) smooth features uniformly across edges, blurring 1-pixel barriers that separate distinct topological chambers.

1.2. The Broader Research Motivation: Structural Computation

Beyond `ARC-AGI-2` itself, our work is motivated by a broader foundational problem that we term **Structural Computation** (building upon the relational inductive bias framework of Battaglia et al., 2018, the Geometric Deep Learning program of Bronstein et al., 2021, and cellular sheaf theory from Hansen & Ghrist, 2019, and Bodnar et al., 2022):

> **Central Problem of Structural Computation:** *How can we construct compact, end-to-end differentiable computational substrates in which the structural properties of information—domain boundaries, cardinal conservation quotas, disentangled symbol-geometry fibers, non-commutative directional relationships, local-to-global topological consistency, associative memory, and complexity-bounded state adaptation—participate directly in the mathematical operations of the network, rather than having to be statistically reconstructed from unstructured latent vectors?*

As highlighted by constructive discussions with top ARC Prize 2026 competitors (including Bushi Liu, Yi-Chia Chen, and Chan Kha Vu), solving the full spectrum of human-like abstract reasoning ultimately requires **both** broad world knowledge (physical intuitions about gravity, lasers, or fluid flow, which large pretrained LLMs or video/JEPA world models acquire from massive corpora) **and** a compact, structure-preserving geometric substrate (a "geometric visual cortex" that can manipulate boundaries, symmetries, and topological constraints natively in weight space without brute-forcing thousands of text tokens or Python scripts).

In this monograph, we use `ARC-AGI-2` as a concrete experimental testbed to investigate the geometric-substrate side of this problem in isolation—training a compact **6.57M-parameter** network (`FGNS-TRM v5.0.0`) from scratch on a single consumer GPU (`NVIDIA RTX 5070`, 12 GB VRAM) with zero LLMs and zero symbolic DSL search. We present the complete mathematical formulation of the architecture, an empirical audit of all figures and ablations, and a candid diagnostic analysis of prototype limitations and future directions.

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2. Related Work & Taxonomy of Contributions

2.1. Prior Paradigms in Compact Neural Reasoning

Our architecture builds directly upon four foundational lines of research:

1. **Dual-State Recursive Refinement (`TRM`):** Jolicoeur-Martineau (2025) demonstrated in *Less is More: Recursive Reasoning with Tiny Networks* (`TRM`) that maintaining separate latent reasoning ("thought") and answer-draft states across recurrent iterations enables compact networks (~7M parameters) to refine solutions progressively. We adopt dual-state recurrence $(Z^{(\tau)}, P^{(\tau)})$ across $N = 4$ steps, replacing generic MLP/Transformer updates with proper-time dilated Cauchy resolvents and Lyapunov-gated support-residual corrections.
2. **Closed-Form Continuous-Time Resolvents (`CfC`):** Hasani et al. (2022) showed that liquid time-constant differential equations can be integrated in closed form via exponential Cauchy kernels, avoiding stiff numerical ODE solvers. We extend this principle to 2D spatial fields with boundary-viscosity proper-time dilation (`ACCROFlowEngine`).
3. **Minimum Description Length Compression (`CompressARC`):** Liao & Gu (2025) framed single-task ARC inference as MDL compression without pretraining. We incorporate MDL-inspired complexity regularization into transductive test-time adaptation (`LOO-TTT`) by penalizing task-specific operator weight perturbations $\Delta W = W - W_{\text{init}}$ under a cuSOLVER-free Newton-Schulz Schatten-1 nuclear norm.
4. **Cellular Sheaf Diffusion & Frame Bundles:** Hansen & Ghrist (2019) and Bodnar et al. (2022) introduced cellular sheaf Laplacians on graphs, where nodes communicate through restriction maps into edge stalks. We lift cellular sheaves onto the compact Stiefel manifold $\mathcal{V}_{32}(\mathbb{R}^{128})$ via 5th-order Newton-Schulz orthogonalization and couple them with anisotropic Perona-Malik-Randers edge-stopping barriers.

2.2. Taxonomy of Contributions Across Five Pillars of Structural Computation

To maintain strict epistemic clarity, **Table 1A** organizes the components of `FGNS-TRM v5.0.0` across five functional pillars of Structural Computation, explicitly distinguishing established prior foundations, our implemented architectural mechanisms, hand-engineered analytical basis relaxations, and untested research hypotheses.

Functional Pillar of Structural Computation	Prior Foundations	Implemented Mechanisms in `FGNS-TRM v5.0.0`	Epistemic Status & Empirical Verification
I. Structural Constraints *(Bounding output geometry & palette quotas upstream)*	Woodbury-Tikhonov ridge regression; Hamilton-Hare apportionment (`LRQ`)	**Stage I** `BoundaryMetricSizePredictor` (`Method 1`) & **Stage II** `EquivariantPaletteHistogramPredictor` with $> 80\%$ diode sink & soft Potts chemical bias	**Empirically Strong:** `76.74%` (`132/172`) exact size on 120 Public Eval tasks (`88.0%`, `0.40 px` MAE on $N = 25$ ablation suite); `-59.1%` palette EMD (`1.3488` → `0.5523`) and `48.0%` exact budget match. *Note: 1-Jet basis $\phi(X)$ and diode threshold are hand-engineered priors.*
II. Disentangled Geometric Representations *(Separating symbol identity from spatial group actions)*	Gram-Schmidt orthogonalization; Rodrigues SO(3) formula; Randers metrics	**Two-pass `MGS2` Orthogonal Split** $\mathbb{R}^{128} = \mathcal{H}^{(11)}_{\text{color}} \oplus \mathcal{H}^{(117)}_{\text{geom}}$ (`Lemma 1`) + 64-byte **`NA3D` 3D simplices** $(\mathbf{L}, \mathbf{C}, \mathbf{R}, \beta)$ & S^9 fiber readout	**Proven Linear Invariance (`Lemma 1`) & Empirically Strong 3D Frame:** 3D `NA3D` yields **+6.54% pixel acc** (`73.10%` vs. `66.56%`) over flat 0D ViT tokens; adding `MGS2` split reduces rotational color leakage from `0.7881` to `1.05e-7` ($N = 25$) and `0.2739` to `1.54e-8` on patch tests (`100%` vs. `28.9%` patch acc) while keeping aggregate 1-epoch accuracy at `73.10%` vs. `72.98%` (`+0.12` pp).
III. Local-to-Global Propagation & Topology *(Directional rays, boundary walls & non-local Green solve)*	State-space scans (`Mamba-2`); Cellular sheaves; Implicit differentiation	**8-Axis `OctoWave` CUDA C++17 SSM** with Householder bounce + **Stiefel Sheaves** (`Lemma 2`) + 23-channel **`SpectralGreenWoodburyFunction`**	**Verified Operator Invariants, Hybrid Basis:** `11.3104` post-`RMSNorm` diagonal impulse-response norm (vs. `0.0000` 2-axis); `0.707` Dirichlet ratio (vs. `7.840` anti-diff); `7.76e-10` FP64 `gradcheck` error. *Note: 23-channel basis $\Psi(X)$ was engineered using Public Eval archetypes; mixer pixel-acc margins are `~0.05%-0.9%` at 1 epoch.*
IV. Associative Memory & State Dynamics *(Rule transport & iterative relaxation)*	Dual-state recurrence (`TRM`); Cauchy resolvents (`CfC`)	**LOO Holographic Support Tape** with SO(3) parallel transport + **J-Involutive Antisymmetric Interaction (`True XOR`)** + **4-Step Dual-State Cauchy Decoder**	**Empirically Strong Recurrence:** 4-step Cauchy recurrence yields **+5.66% pixel accuracy** (`73.11%` vs. `67.45%` for $T = 1$) and `90.7%` monotonic step descent (`32.9%` → `73.3%` → `100%` → `100%` crystallization on `13e47133`).
V. Complexity-Controlled Adaptation *(Task adaptation without coordinate memorization)*	LoRA (`Hu et al., 2021`); Schatten norms; $D_4 \times S_9$ augmentations	**Geodesic** $\mathfrak{so}(3)^8$ **+ MDL-Inspired Schatten-1 `LOO-TTT`** (`Method 2`) via coupled Newton-Schulz iterations	**Verified Rank Compression & Group Equivariance:** Compresses participation effective rank r_{eff} : 3.46 → 1.10; reduces 5-task 8-step query loss `0.4981` → `0.4736` (vs. `0.5126` degradation for Full-FT); `100.0%` vs. `40.74%` on $D_4 \times S_9$ stress test.

Table 1A | Taxonomy of contributions in `FGNS-TRM v5.0.0` across the five pillars of Structural Computation, separating proven algebraic lemmas, empirically validated gains, hybrid hand-engineered bases, and open research directions.

3. Formal Problem Formulation & Exact 6.57M-Parameter Audit

3.1. Transductive Few-Shot Tensor Formulation & Static Support Horizon ($K_{\text{static}} = 6$)

An `ARC-AGI-2` task $\mathcal{T} = (\mathcal{D}_{\text{sup}}, \mathcal{D}_{\text{query}})$ provides K demonstration pairs $\mathcal{D}_{\text{sup}} = \{(X_k^{\text{sup}}, Y_k^{\text{sup}})\}_{k=1}^{K}$ and one or more test query inputs $X_q \in \Omega_{H_{\text{in}} \times W_{\text{in}}}(\mathbb{Z}_{10})$, where each grid has dimensions $1 \le H, W \le 30$ and integer symbols $c \in \{0, \dots, 9\}$.

To eliminate dynamic memory fragmentation in `CUDACachingAllocator` during batched meta-training and test-time adaptation, `FastVectorizedBatcher` (`fgns/env/batcher.py`) embeds every grid into a canonical 30×30 buffer over the augmented 11-symbol vocabulary $\Sigma_{11} = \{0, \dots, 9\} \cup \{\text{PAD} = 10\}$ accompanied by a boolean validity mask $M(y, x) \in \{0, 1\}^{30 \times 30}$, and buckets the demonstration horizon to a static capacity $K_{\text{static}} = 6$:

- If a task has $K < 6$ demonstration pairs, the remaining $6 - K$ support slots are filled with `PAD = 10` (`in_masks = False`) and masked out via `pair_valid_mask` ($\mathbf{v}_{\text{pair}} \in \{0, 1\}^{B \times 6}$) so padded slots contribute zero weight to all attention, Gram, and loss reductions.
- If a task has $K > 6$ demonstration pairs (e.g., $K \in \{7, 8\}$), `collate_task_pairs` slices the first $K_{\text{static}} = 6$ pairs (`support_pairs[:k_eff]`); during training and `LOO-TTT`, demonstration pairs are randomly permuted across steps so all demonstrations are sampled across the optimization trajectory.

3.2. Verified Parameter Audit (`6,574,411` Parameters)

Every submodule in `FinslerGaugeSheafNetwork` (`fgns/models/fgns_network.py`) is aligned to $d_{\text{model}} = 128$ channels (divisible by 32 for Tensor Core GEMMs and partitioned into $C = 8$ `NA3D` cells of stride 16). **Table 1B** reports the exact parameter count of every top-level child module audited directly from `model_final.pt`.

Architectural Stage & Submodule (`FinslerGaugeSheafNetwork`)	Primary Mathematical Role	Key Tensor Dimensions	Exact Parameters	Share (%)
1. `slot_binder` (`RelationalSlotBinder`)	Betti-0 spectral slot seeding, 12-slot competitive attention, directed hypergraph & seed-anchored bipartite morphism	$S = 12,\ d_{\text{model}} = 128,\ d_{\text{rel}} = 32$	`1,096,026`	`16.67%`
2. `lie_extractor` (`LieAlgebraTransitionExtractor`)	Support pre-image alignment, Cartan decomposition $\mathfrak{gl}(128) = \mathrm{Sym}(128) \oplus \mathfrak{so}(128)$, Rayleigh power iteration & Lorentz $\mathbb{H}^{128}_{\kappa}$ rule lift	$D = 128,\ \mathbb{H}^{128}_{\kappa} \subset \mathbb{R}^{129}$	`922,757`	`14.04%`
3. `accro_decoder` (`DualStateCauchyRecurrentDecoder`)	4-step $(Z^{(\tau)}, P^{(\tau)})$ Cauchy resolvent flow, 6-stream `ForceSynthesizer`, `SpatialMooreSheafDiffusion` & S^9 Potts readout	$N = 4,\ d_{\text{edge}} = 32,\ V = 11$	`848,231`	`12.90%`
4. `octowave_ssm` (`OctoWaveSSM`, 2 blocks)	8-compass line-parallel CUDA C++17 state-space scan, Hodge boundary normal guidance & single-bounce Householder reflection	2 layers, 8 axes, $D = 128$	`816,440`	`12.42%`
5. `projector` (`TopologicalCoordinateEncoder`)	Orthonormal palette $\mathcal{V}_{11}(\mathbb{R}^{128})$, Chebyshev graph Laplacian, multiscale Kronecker-Lie resonator & `NA3D` cell synthesis	$V = 11,\ D = 128,\ 8 \text{ cells} \times 16$	`708,429`	`10.78%`
6. `holographic_tape` (`SupportCrossAttentionMemory`)	Dual-scale (5×5 local glyph + global centroid) support indexing, `NA3D` chiral gauge attention & SO(3) Rodrigues parallel transport	$K_{\text{static}} = 6,\ H = 4 \text{ heads},\ D = 128$	`626,820`	`9.53%`
7. `size_oracle` (`BoundaryMetricSizePredictor`)	**Stage I Prior:** Bochner harmonic fiber, envelope pooler, Cartan rule modulator & coupled Boltzmann-Woodbury 1-Jet Stokes solver	$\mathbb{Z}_{61} \times \mathbb{Z}_{61},\ d_{\text{fiber}} = 32$	`625,184`	`9.51%`
8. `slot_sheaf` (`CellularSheafOperator`)	Stiefel $\mathcal{V}_{32}(\mathbb{R}^{128})$ 5th-order Newton-Schulz restriction maps, coboundary Laplacian & Woodbury-Schur resolvent	$S = 12,\ d_{\text{node}} = 128,\ d_{\text{edge}} = 32$	`540,928`	`8.23%`
9. `budget_oracle` + `spectral_green_operator` + **Routers**	**Stage II Prior** (`320,599`), `query_canvas_initializer` (`32,768`), `chamber_fold_proj` (`16,384`), `sinkhorn_projector` (`15,619`), `na3d_rotor_proj` (`3,072`), `pullback_router` (`1,032`), `spectral_green_operator` (`239`) & gates	$C = 10,\ K_{\text{basis}} = 23,\ \mathcal{B}_{11}$	`389,596`	`5.93%`
TOTAL (`FinslerGaugeSheafNetwork` v5.0.0)	**Complete Standalone Continuous Geometric-Analytical Engine**	**25.08 MB (FP32) / 12.54 MB (BF16)**	`6,574,411`	`100.00%`

Table 1B | Exact parameter breakdown across all submodules of `FinslerGaugeSheafNetwork` (`fgns/models/fgns_network.py`), verified against `artifacts/checkpoints/model_final.pt`.

4. Two-Stage Upstream Structural Priors: Bounding Dimensions and Palette Quotas

4.1. Information-Theoretic Motivation: Contracting Combinatorial Phase-Space Cardinality

Consider the number of discrete grid configurations admissible under three successive levels of structural conditioning:

1. **Unconstrained Padded Lattice:** Over the 30×30 buffer with 11 states per cell, the raw output space contains $|\Omega_{\text{raw}}| = 11^{900} \approx 1.79 \times 10^{937}$ configurations.
2. **Stage I Dimension Conditioning:** Once target dimensions $(H^\star, W^\star) \in \{1, \dots, 30\}^2$ are resolved (`BoundaryMetricSizePredictor`), all $900 - K_{\text{cap}}$ exterior cells ($K_{\text{cap}} = H^\star W^\star$) are deterministically clamped to `PAD=10`, while the K_{cap} interior cells are restricted to domain colors $\{0, \dots, 9\}$, reducing the number of admissible interior states to $|\Omega_{\text{Stage I}}| = 10^{K_{\text{cap}}}$.
3. **Stage II Multinomial Quota Conditioning:** If the exact integer color counts $\mathbf{N}^\star = (N_0^\star, \dots, N_9^\star) \in \mathbb{N}_0^{10}$ satisfying $\sum_{c=0}^{9} N_c^\star = K_{\text{cap}}$ are known, the number of distinct spatial arrangements matching that histogram is given by the multinomial coefficient $|\Omega_{\text{Stage II}}| = \frac{K_{\text{cap}}!}{\prod_{c=0}^{9}(N_c^\star)!}$.

On Task `0934a4d8` ($30 \times 30 \to 9 \times 3$, $K_{\text{cap}} = 27$, with ground-truth counts $N_1 = 3, N_2 = 2, N_4 = 4, N_6 = 6, N_7 = 8, N_9 = 4$), the theoretical combinatorial cardinality ratio between $|\Omega_{\text{raw}}|$ and $|\Omega_{\text{Stage II}}|$ exceeds 10^{920}. Importantly, this ratio represents a **theoretical measure of combinatorial state-space cardinality**, not a wall-clock runtime speedup factor; furthermore, in `FGNS-TRM v5.0.0`, Stage II quotas are applied as a **soft Potts chemical potential bias** rather than a hard combinatorial permutation lock, allowing local spatial operators to refine imperfect quota estimates during decoding.

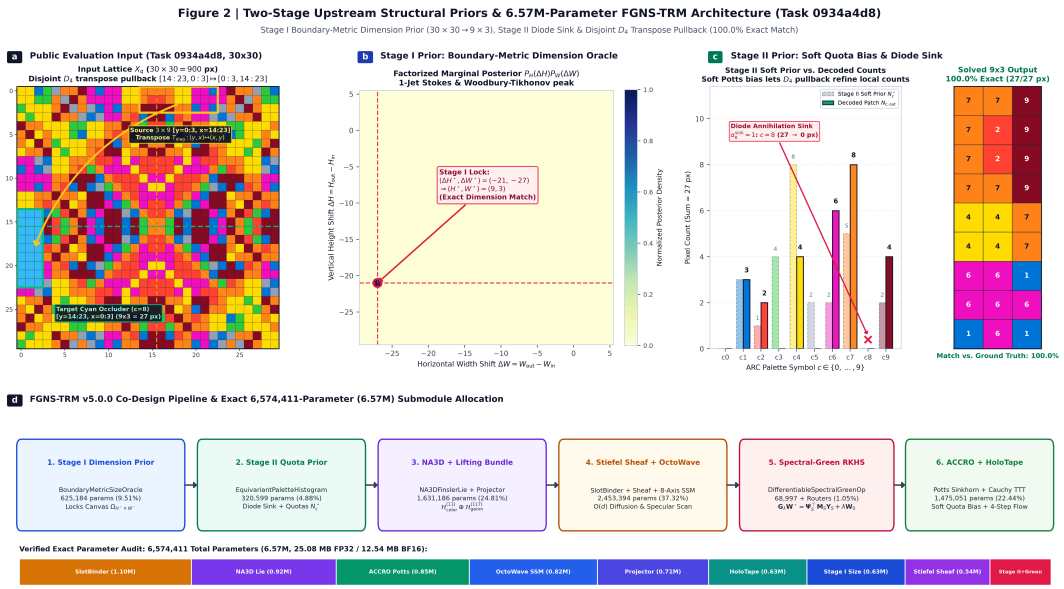


Figure 2 | Empirical demonstration of the Two-Stage Upstream Structural Priors on ARC-AGI-2 Public Evaluation Task `0934a4d8` ($30 \times 30 \to 9 \times 3$, `0` TTT steps): (a) Input lattice X_q showing the 9×3 cyan occluder `[14:23, 0:3]` ($c = 8$, 27 px) and the disjoint 3×9 top source `[0:3, 14:23]` selected via main-diagonal transposition $T_{\text{diag}} : (y, x) \mapsto (x, y)$ because the vertical mirror image across $y_s = 15.5$ self-overlaps with the occluder on rows `14..17`; (b) Stage I factorized marginal displacement posterior $P_H(\Delta H)P_W(\Delta W)$ locking $(\Delta H^\star, \Delta W^\star) = (-21, -27) \implies (H^\star, W^\star) = (9, 3)$; (c) Stage II soft Hamilton-Hare prior quotas $N_c^\star$ (`[0, 3, 1, 4, 8, 2, 2, 5, 0, 2]`, dashed bars, with diode sink $\alpha_8^{\text{sink}} = 1 \implies N_8^\star = 0$) versus final decoded pixel counts (`[0, 3, 2, 0, 4, 0, 6, 8, 0, 4]`, solid bars, 100.0% exact match `27/27 px`); and (d) Exact `6,574,411`-parameter allocation across submodules.

4.2. Stage I: Coupled Boltzmann-Gibbs & Woodbury-Tikhonov 1-Jet Dimension Estimator (`Method 1`)

To predict $(H^\star, W^\star)$ without relying on unconstrained global average pooling, `BoundaryMetricSizePredictor` (`fgns/models/size_predictor/boundary_size_predictor.py`) combines a Shared-SRAM CUDA flood-fill and optical ray kernel (`_finsler_chordal_cavity_sram_kernel` in `fgns/kernels/cuda_fgns_ops.py`) with:

1. **Bochner Multi-Harmonic Scale Fiber (`HarmonicGeometricFiber`):** Encodes input dimensions $(H_{\text{in}}, W_{\text{in}})$ via multi-frequency Fourier harmonics on $\mathbb{T}^2$ and five $\mathrm{Sim}(2)$ conformal invariants: log-area $\ln(H_{\text{in}} W_{\text{in}})$, log-aspect $\ln(H_{\text{in}}/W_{\text{in}})$, normalized diagonal, orientation angle $\arctan(H_{\text{in}}/W_{\text{in}})$, and the Minkowski isoperimetric quotient $Q_{\text{iso}} = \frac{\pi H_{\text{in}} W_{\text{in}}}{(H_{\text{in}}+W_{\text{in}})^2} \leq \frac{\pi}{4}$.
2. **Coordinate-Directed Envelope Pooler & Cartan Rule Modulator:** `SpatialEnvelopeAttentionPooler` probes the latent feature field using a 6-channel metric coordinate bundle, and `CartanRuleModulator` deforms the pooled geometry via the bounded affine group action $\mathbf{z}_{\text{mod}} = \mathrm{RMSNorm}\big(\mathbf{z}_{\text{geo}} \odot (1 + \tanh \boldsymbol{\gamma}(\mathbf{z}_{\text{rule}})) + \boldsymbol{\beta}(\mathbf{z}_{\text{rule}})\big)$.
3. **8-Dimensional 1-Jet Geometric Basis $\boldsymbol{\phi}(X) \in \mathbb{R}^8$:** Extracted along each spatial axis ($L \in \{H, W\}$, $L^\perp \in \{W, H\}$):

$$\boldsymbol{\phi}_L(X) = \Big[\underbrace{L_{\text{in}}(X)}_{\phi_0\text{: Active Span}}, \underbrace{L_{\text{bbox}}(X)}_{\phi_1\text{: FG BBox}}, \underbrace{L_{\text{cham}}(X)}_{\phi_2\text{: Chamber Span}}, \underbrace{L_{\text{occ}}(X)}_{\phi_3\text{: Vanishing Occluder}}, \underbrace{L^\perp_{\text{in}}(X)}_{\phi_4\text{: Transpose Span}}, \underbrace{L^\perp_{\text{occ}}(X)}_{\phi_5\text{: Transpose Occluder}}, \underbrace{L_{\text{max-col}}(X)}_{\phi_6\text{: Max-Color BBox}}, \underbrace{L_{\text{min-col}}(X)}_{\phi_7\text{: Min-Color BBox}} \Big]^\top$$

Method 1 (Coupled Boltzmann-Gibbs & Woodbury-Tikhonov 1-Jet Boundary Dimension Estimator). *Given valid support demonstrations $\{(X_k, Y_k)\}_{k=1}^{K}$ with conjunctive validity weights $w_k = v_k / \sum_j v_j$, Stage I evaluates 10 discrete geometric law hypotheses $\hat{L}_{q,m}$ ($m \in \{1, \dots, 10\}$: additive shift, multiplicative homothety, constant extent, foreground bounding box + offset, divider chamber span, vanishing occluder box, transposed occluder box, extremal largest/smallest single-color object boxes, and non-square transposition) alongside a continuous 11th hypothesis ($m = 11$) obtained by solving the dual $K \times K$ Woodbury-Tikhonov Gram system over the 8-channel feature map $\boldsymbol{\psi}(X) \in \mathbb{R}^8$:*

$$\mathbf{w}_{\text{Woodbury}} = (\boldsymbol{\Psi}_S \odot \mathbf{g}_{\text{Fisher}})^\top (\mathbf{G}_K + \lambda_{\text{Tik}} I_K)^{-1} \boldsymbol{\Delta}_S, \qquad \hat{L}_{q,11} = \text{clip}\big(L_{\text{in}}(X_q) + \text{round}\big(\boldsymbol{\psi}(X_q)^\top \mathbf{w}_{\text{Woodbury}}\big),\, 1,\, 30\big)$$

where $\mathbf{G}_K = \frac{1}{2}\big((\boldsymbol{\Psi}_S \odot \mathbf{g}_{\text{Fisher}})\boldsymbol{\Psi}_S^\top + \boldsymbol{\Psi}_S(\boldsymbol{\Psi}_S \odot \mathbf{g}_{\text{Fisher}})^\top\big) \in \mathbb{R}^{K \times K}$, $\lambda_{\text{Tik}} = 10^{-3}$, and $(\mathbf{G}_K + \lambda_{\text{Tik}} I_K)^{-1}$ is evaluated via 6 Newton-Schulz iterations in FP32. Each hypothesis $m \in \{1, \dots, 11\}$ is scored by its weighted support residual $\mathcal{E}_m = \sum_{k=1}^{K} w_k |L_{\text{out}}(Y_k) - \hat{L}_{k,m}| + \delta_{\text{Occam},m}$ ($\delta_{\text{Occam},m} \in [0, 0.06]$) and votes at its integer shift index $c_m(X_q) = 30 + (\hat{L}_{q,m} - L_{\text{in}}(X_q)) \in [0, 60]$ via a Boltzmann-Gibbs Gaussian mixture:

$$\pi_m = \frac{\exp(-18.0(\mathcal{E}_m - \mathcal{E}_{\min}))}{\sum_{j=1}^{11} \exp(-18.0(\mathcal{E}_j - \mathcal{E}_{\min}))}, \qquad \ell_{\text{prior}}(c) = \sum_{m=1}^{11} \pi_m \exp\left(-\frac{(c - c_m(X_q))^2}{2(0.50)^2}\right), \quad c \in \{0, \dots, 60\}$$

Crucially, evaluating each law at its discrete peak $c_m(X_q)$ in logit space before softmax prevents multi-law averaging blur when two laws coincide on training pairs (e.g., when a vanishing occluder in a demonstration happens to have the same width as a divider chamber). After the 4-step recurrent decoder produces the terminal carrier log-odds field $\psi(y, x)$, `refine_with_terminal_carrier_mask` applies a residual Stokes cardinality correction gated by $\tanh(2.5\,\mathcal{E}_{\min})$, leaving zero-error analytical locks ($\mathcal{E}_{\min} = 0$) unperturbed.

4.3. Stage II: S_{10}-Equivariant Role Descriptors, Diode Annihilation Sink & Soft Hamilton-Hare Quotas

Conditioned on the Stage I capacity $K_{\text{cap}} = H^\star \times W^\star$, `EquivariantPaletteHistogramPredictor` (`fgns/models/palette_histogram/histogram_predictor.py`) estimates the output color distribution without relying on raw color indices:

1. **S_{10}-Equivariant 8D Structural Role Descriptors:** For every domain color $c \in \{0, \dots, 9\}$, `extract_color_structural_roles` (`fgns/kernels/spectral_laplacian.py`) computes an 8-dimensional geometric role vector $\boldsymbol{\rho}_c(X) = [\rho_{c,1}, \dots, \rho_{c,8}]^\top \in [0, 1]^8$ measuring: (1) active area density, (2) canvas border fraction, (3) full-span divider-line coverage, (4) bounding-box rectangularity fill $M_c / (H_{B_c} W_{B_c})$ (identifying solid rectangular masks such as $c = 8$ in `0934a4d8`), (5) isolated 1×1 seed singleton ratio, (6) multi-pixel structural wall ratio, (7) enclosed Dirichlet cavity adjacency, and (8) bounding-box aspect ratio $\min(H_{B_c}, W_{B_c}) / \max(H_{B_c}, W_{B_c})$. Because $\boldsymbol{\rho}_c(X)$ is computed identically for each color and processed through shared per-color weights (`per_color_equivariant_mlp` and `role_equivariant_proj`), the role representation is equivariant to any permutation $\pi \in S_{10}$ of the 10 domain colors.
2. **Diode Annihilation Sink ($\alpha_c^{\text{sink}} \in \{0, 1\}$):** In tasks with temporary auxiliary markers, occluders, or construction lines, a foreground color $c \in \{1, \dots, 9\}$ present in the inputs vanishes completely in the demonstration outputs. We measure the support disappearance rate:

   $$r_c^{\text{vanish}} = \frac{\sum_{k=1}^{K} \mathbb{I}[N_c(X_k) > 0 \ \wedge \ N_c(Y_k) = 0]}{\max\left(1, \sum_{k=1}^{K} \mathbb{I}[N_c(X_k) > 0]\right)}, \qquad \alpha_c^{\text{sink}} = \mathbb{I}\big[r_c^{\text{vanish}} > 0.80\big] \quad (c \geq 1,\ \alpha_0^{\text{sink}} \equiv 0)$$

   When $\alpha_c^{\text{sink}} = 1$, the diode sink zeroes the active-color mask for color c in the simplex prior and adds a -8.0 column barrier in the log-domain Sinkhorn-Birkhoff color transition projector (`LogDomainSinkhornBirkhoffProjector`), preventing vanished mask colors from leaking into the output palette.
3. **Dual-Gram Woodbury-Tikhonov Simplex Prior & Sherman-Morrison LOO Leverage:** Over the support simplex basis $\boldsymbol{\Phi}_S \in \mathbb{R}^{B \times K \times M}$ (encoding constant offset, input proportions $\mathbf{p}_{\text{in}}$, area-rescaled proportions, foreground-conserving proportions, and 3-bin cavity mass), `compute_dual_gram_tikhonov_resolvent` solves for the simplex transition operator $\mathbf{W}_{\text{op}} = \boldsymbol{\Phi}_{S,w}^\top (\mathbf{G}_{\text{sym}} + \lambda_{\text{Tik}} I_K)^{-1} \Delta\mathbf{P}_{S,w}$ via 6 Newton-Schulz iterations and evaluates the exact Leave-One-Out residual via the **Sherman-Morrison-Woodbury diagonal hat-matrix identity** $\mathbf{r}_{\text{LOO}} = \frac{|\Delta\mathbf{P}_{S,w} - \mathbf{H}\Delta\mathbf{P}_{S,w}|}{\max(0.20, 1 - \text{diag}(\mathbf{H}))}$.
4. **Straight-Through Hamilton-Hare Apportionment (`StraightThroughHareQuota`) & Soft Potts Bias:** Given the refined simplex proportions $\mathbf{p}^\star \in \Delta^9$, `differentiable_hare_lrq` (`fgns/primitives/hare_lrq.py`) projects $\mathbf{p}^\star K_{\text{cap}}$ onto exact integers $\mathbf{N}^\star \in \mathbb{N}_0^{10}$ in the forward pass using vectorized double-`argsort` 0-indexed remainder ranking

$\mathrm{rank}_0(r_c) \in \{0,\dots,9\}$ (where $r_c = p_c^\star K_{\text{cap}} - \lfloor p_c^\star K_{\text{cap}} \rfloor$):

$$N_c^\star = \lfloor p_c^\star K_{\text{cap}} \rfloor + \mathbb{I}\left[\mathrm{rank}_0(r_c) < \mathrm{round}(K_{\text{cap}}) - \sum_{j=0}^{9} \lfloor p_j^\star K_{\text{cap}} \rfloor\right], \qquad \sum_{c=0}^{9} N_c^\star = \mathrm{round}(K_{\text{cap}})$$

and propagates gradients in the backward pass via the bilinear straight-through surrogate $q_c = p_c^\star K_{\text{cap}}$ ($\partial \mathcal{L}/\partial p_c^\star = K_{\text{cap}}\, g_c$, $\partial \mathcal{L}/\partial K_{\text{cap}} = \sum_c p_c^\star\, g_c$). During the 4-step recurrent decoding loop, at each step $\tau \in \{1,\dots,4\}$, `evaluate_dynamic_chemical_biases` compares $N_c^\star$ against the current soft occupancy $\hat{O}_c^{(\tau)} = \sum_{u \in \Omega} P_c^{(\tau)}(u)$ to produce a bounded chemical potential bias:

$$\beta_c^{(\tau)} = g_{\text{chem}} \tanh\left(\frac{N_c^\star - \hat{O}_c^{(\tau)}}{\sqrt{\max(1, K_{\text{cap}})}}\right), \qquad \Delta Z_c^{\text{chem}} = \frac{1}{2}\tanh(g_{\text{logit}})\, \beta_c^{(\tau)}$$

Because $|\Delta Z_c^{\text{chem}}| \le 0.50\, g_{\text{chem}}$, strong local spatial evidence (such as the disjoint D_4 transpose pullback on Task `0934a4d8` in **Figure 2c**) can override minor inaccuracies in the global Stage II quota estimate (`[0, 3, 1, 4, 8, 2, 2, 5, 0, 2]` $\rightarrow$ exact decoded histogram `[0, 3, 2, 0, 4, 0, 6, 8, 0, 4]`), whereas colors annihilated by the diode sink ($\alpha_8^{\text{sink}} = 1$) receive an additional $-\gamma_{\text{alpha}} = -6.0$ alphabet barrier.

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5. Orthogonal Color-Geometry Splitting & Non-Abelian 3D Frames (`NA3D-Finsler-Lie`)

5.1. Two-Pass Modified Gram-Schmidt Splitting (`Lemma 1`)

Let $E = [\mathbf{e}_0,\dots,\mathbf{e}_{10}]^\top \in \mathbb{R}^{11\times 128}$ denote the normalized prototype vectors of the 11 vocabulary symbols (`OrthonormalPaletteFrame` in `fgns/primitives/potts_latching.py`). In `_get_exact_orthonormal_palette_basis` (`fgns/models/fgns_network.py`), we construct a **numerically orthonormal palette basis** $Q = [\mathbf{q}_0,\dots,\mathbf{q}_{10}]^\top \in \mathcal{V}_{11}(\mathbb{R}^{128})$ ($\mathrm{span}(Q) = \mathrm{span}(E)$) in IEEE-754 FP32 using **Two-Pass Modified Gram-Schmidt (`MGS2`)** with `allow_tf32 = False` (achieving measured Gram error $\|QQ^\top - I_{11}\|_{\max} = 2.38 \times 10^{-7}$ without invoking cuSOLVER QR kernels):

$$\mathbf{v}_k^{(1)} = \mathbf{e}_k - \sum_{j=0}^{k-1} \langle \mathbf{e}_k, \mathbf{q}_j \rangle \mathbf{q}_j, \qquad \mathbf{v}_k^{(2)} = \mathbf{v}_k^{(1)} - \sum_{j=0}^{k-1} \langle \mathbf{v}_k^{(1)}, \mathbf{q}_j \rangle \mathbf{q}_j, \qquad \mathbf{q}_k = \frac{\mathbf{v}_k^{(2)}}{\max(\|\mathbf{v}_k^{(2)}\|_2, 10^{-8})}$$

Lemma 1 (Orthogonal Color-Geometry Subspace Invariance under Projected Updates).
Let $Q \in \mathcal{V}_{11}(\mathbb{R}^{128})$ satisfy $QQ^\top = I_{11}$, and define the orthogonal projectors $P_{\text{color}} = Q^\top Q \in \mathbb{R}^{128\times 128}$ and $P_{\text{geom}} = I_{128} - Q^\top Q \in \mathbb{R}^{128\times 128}$. Then $\mathbb{R}^{128}$ decomposes as the orthogonal direct sum:

$$\mathbb{R}^{128} = \mathcal{H}_{\text{color}}^{(11)} \oplus \mathcal{H}_{\text{geom}}^{(117)}, \qquad P_{\text{color}}^2 = P_{\text{color}}, \quad P_{\text{geom}}^2 = P_{\text{geom}}, \quad Q P_{\text{geom}} \equiv \mathbf{0}_{11\times 128}$$

Consequently, for any latent state $\mathbf{h}(u) \in \mathbb{R}^{128}$ and any spatial or Lie-group perturbation $\Delta\mathbf{h}_{\text{raw}}(u) \in \mathbb{R}^{128}$ (such as $\mathrm{SO}(3)^8$ Rodrigues rotations or $\mathfrak{gl}(128)$ transition actions) that is projected via two-pass residual orthogonalization $\Delta\mathbf{h}_\perp(u) = P_{\text{geom}}\big(P_{\text{geom}}\Delta\mathbf{h}_{\text{raw}}(u)\big)$, the updated state $\mathbf{h}'(u) = \mathbf{h}(u) + g_{\text{na3d}}\,\Delta\mathbf{h}_\perp(u)$ identically preserves all inner products in $\mathcal{H}_{\text{color}}^{(11)}$:

$$Q\,\mathbf{h}'(u) = Q\,\mathbf{h}(u) + g_{\text{na3d}} \underbrace{(QP_{\text{geom}})}_{\mathbf{0}} P_{\text{geom}}\Delta\mathbf{h}_{\text{raw}}(u) = Q\,\mathbf{h}(u), \qquad \langle \mathbf{h}'(u) - \mathbf{h}(u),\, \mathbf{e}_c \rangle = 0 \quad \forall\, c \in \{0,\dots,10\}$$

Proof. Since $QQ^\top = I_{11}$, $QP_{\text{geom}} = Q(I_{128} - Q^\top Q) = Q - (QQ^\top)Q = Q - I_{11}Q = \mathbf{0}_{11\times 128}$. Because every prototype $\mathbf{e}_c \in \mathrm{span}(Q)$ can be written as $\mathbf{e}_c = \sum_{k=0}^{10} \alpha_{c,k}\mathbf{q}_k$, it follows that $\langle P_{\text{geom}}\Delta\mathbf{h}, \mathbf{e}_c\rangle = 0$ for all $c \in \{0,\dots,10\}$. In finite IEEE-754 FP32 arithmetic where $\|QQ^\top - I_{11}\|_{\max} = 2.38 \times 10^{-7}$, two-pass residual projection $\Delta\mathbf{h}_\perp = P_{\text{geom}}(P_{\text{geom}}\Delta\mathbf{h}_{\text{raw}})$ suppresses residual floating-point coupling to $\sim 10^{-8}$–10^{-7}. ■

Scope & Re-Projection Requirement: Lemma 1 is an exact linear-algebraic invariance identity for any operator whose update is projected by P_{geom} (such as `_apply_na3d_cell_holonomy_action` and the $\mathfrak{gl}(128)$ branch of `_apply_lie_and_birkhoff_operators`). When subsequent modules apply unconstrained dense linear layers (`channel_mixer`, `SwiGLU` FFNs) across all 128 channels, $\mathcal{H}_{\text{color}}^{(11)}$ and $\mathcal{H}_{\text{geom}}^{(117)}$ can re-couple unless explicitly re-projected; only intentional palette permutation operators ($P_{\text{Birkhoff}} - I_{11}$) are designed to act directly on $\mathcal{H}_{\text{color}}^{(11)}$.

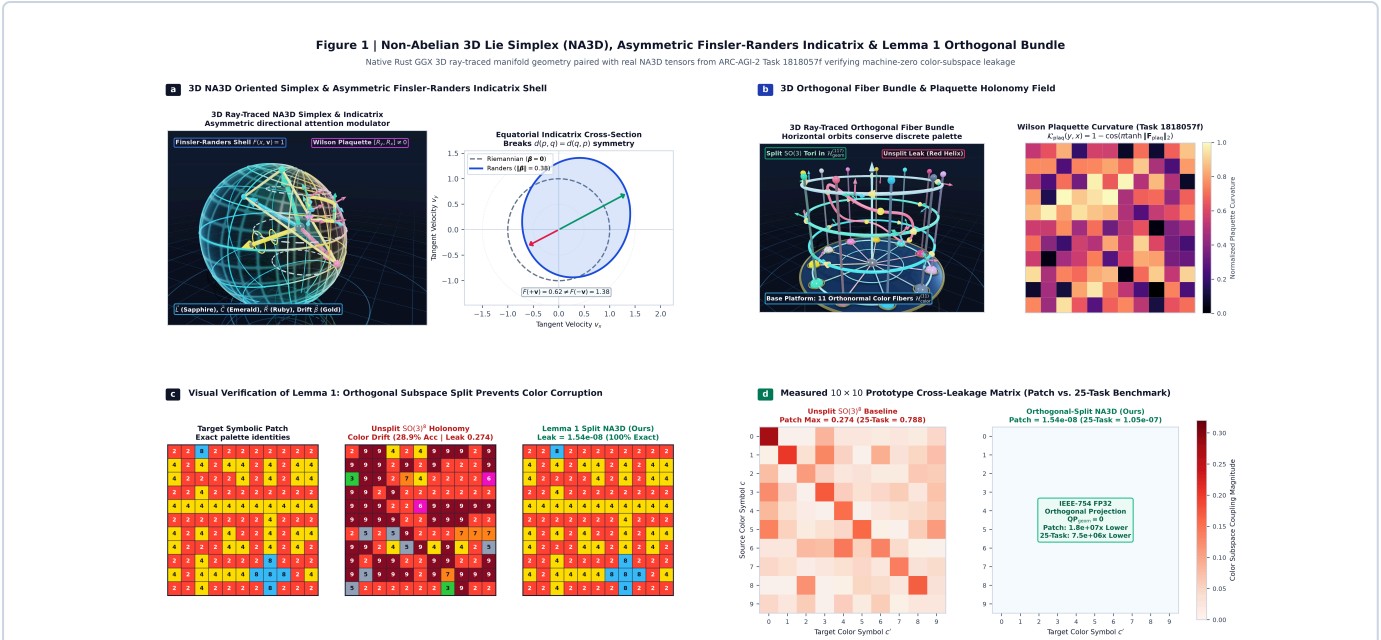


Figure 1 | Geometric structure and empirical verification of `NA3D-Finsler-Lie` and `Lemma 1`: (a) Left: Native Rust Cook-Torrance GGX 3D ray-traced rendering of an `NA3D` oriented simplex $(\mathbf{L}, \mathbf{C}, \mathbf{R})$ and drift covector $\boldsymbol{\beta}$ inside the cutaway Finsler-Randers indicatrix shell $F(x, \mathbf{v}) = 1$, alongside a 2×2 Wilson plaquette loop. Right: Equatorial cross-section showing directional cost asymmetry $F(+\mathbf{v}) = 0.62 \neq F(-\mathbf{v}) = 1.38$. (b) Left: 3D ray-traced orthogonal fiber bundle $\mathbb{R}^{128} = \mathcal{H}^{(11)}_{\text{color}} \oplus \mathcal{H}^{(117)}_{\text{geom}}$ contrasting horizontal gauge orbits in $\mathcal{H}^{(117)}_{\text{geom}}$ against a tilted unsplit leakage helix. Right: Authentic 2×2 Yang-Mills-Wilson plaquette curvature $\mathcal{K}_{\text{plaq}}(y, x)$ extracted from Task `1818057f` (`[y=8..18, x=10..20]`). (c) Visual verification on the 11×11 patch of Task `1818057f`: unsplit $\mathrm{SO}(3)^8$ holonomy corrupts palette prototypes (`28.9%` patch accuracy, white boxes), whereas `Lemma 1` MGS2-split `NA3D` preserves `100.0%` (`121/121 px`) exact palette identities. (d) Measured 10×10 prototype cross-leakage matrix $|\langle \mathbf{h}_{\text{after}}(\mathbf{e}_c) - \mathbf{h}_{\text{before}}(\mathbf{e}_c), \mathbf{e}_{c'} \rangle|$, showing reduction from `0.2739` to `1.54e-8` on the unit-prototype patch (`1.8e7x` lower) and from `0.7881` to `1.05e-7` on the 25-task benchmark (`7.5e6x` lower).

5.2. Non-Abelian 3D Token (`NA3D`) Cells, Finsler-Randers Proper Time & Wilson Plaquette Holonomy

Within the 128-dimensional latent vector (and inside the projected geometric complement $\mathcal{H}^{(117)}_{\text{geom}}$ during holonomy updates), features are partitioned into $C = 8$ `NA3D` cells of 16 FP32 channels each (64 bytes per cell, aligned to GPU cache lines; `NA3DFinslerLieEngine` in `fgns/primitives/stiefel_sheaf.py`):

$$\mathcal{Z}_u^{(c)} = \Big(\underbrace{\mathbf{L}_u^{(c)} \in \mathbb{R}^3}_{\text{Receptor Face}}, \underbrace{\mathbf{C}_u^{(c)} \in \mathbb{R}^3}_{\text{Curvature Core}}, \underbrace{\mathbf{R}_u^{(c)} \in \mathbb{R}^3}_{\text{Emitter Face}}, \underbrace{\boldsymbol{\beta}_u^{(c)} \in \mathbb{B}_1^3}_{\text{Randers Drift}}, \underbrace{\mathbf{m}_u^{(c)} \in \mathbb{R}^4}_{\text{Chiral/Wilson Meta}} \Big), \quad c \in \{1, \dots, 8\}$$

1. **Oriented Chiral Volume & Local Spin:** Each cell spans a signed 3D volume $V_{\text{chiral}}(u) = \tanh\big(\det[\mathbf{L}_u, \mathbf{C}_u, \mathbf{R}_u]\big) = \tanh\big(\langle \mathbf{L}_u, \mathbf{C}_u \times \mathbf{R}_u \rangle\big)$ (which reverses sign under spatial reflections) and an intrinsic spin vector $\mathbf{s}_u = \mathbf{L}_u \times \mathbf{R}_u \in \mathfrak{so}(3)$.
2. **Asymmetric Finsler-Randers Proper-Time Conductance:** For a general Finsler-Randers metric $F(x, \mathbf{v}) = \sqrt{\mathbf{v}^\top A(x) \mathbf{v}} + \langle \boldsymbol{\beta}_x, \mathbf{v} \rangle$ with symmetric positive-definite $A(x) \succ 0$, strict positivity and strong convexity of the indicatrix require the dual Riemannian norm condition $\sqrt{\boldsymbol{\beta}_x^\top A(x)^{-1} \boldsymbol{\beta}_x} < 1$. In our implementation where the local spatial metric is Euclidean ($A(x) = I_3$), this condition simplifies to $\|\boldsymbol{\beta}_u\|_2 < 1$, which we enforce analytically by smoothly retracting the raw drift vector $\mathbf{v}_\beta \in \mathbb{R}^3$ into the open unit ball via $\boldsymbol{\beta}_u = \dfrac{\tanh(\|\mathbf{v}_\beta\|_2)}{\sqrt{\|\mathbf{v}_\beta\|_2^2 + \epsilon}} \mathbf{v}_\beta$ ($\|\boldsymbol{\beta}_u\|_2 < 1$). Across a directed cardinal edge $u \to v$, the interface displacement $\boldsymbol{\delta}_{uv} = \mathbf{L}_v - \mathbf{R}_u$ experiences an asymmetric proper-time delay τ_{uv} and closed-form `CfC` conductance weight w_{uv}:

$$\tau_{uv} = \max\left(10^{-4}, \sqrt{\|\boldsymbol{\delta}_{uv}\|_2^2 + \mu_0^2} + \langle \boldsymbol{\beta}_u - \boldsymbol{\beta}_v, \boldsymbol{\delta}_{uv} \rangle\right), \qquad \sigma_{uv} = \exp\left(-\frac{\tau_{uv}}{\tau_0}\right), \qquad w_{uv} = \frac{\sigma_{uv}}{\sum_{v' \in \mathcal{N}_4(u)} \sigma_{uv'}}$$

   Because $\langle \boldsymbol{\beta}_u - \boldsymbol{\beta}_v, \boldsymbol{\delta}_{uv} \rangle$ changes sign when edge orientation reverses, $\tau_{uv} \neq \tau_{vu}$, acting as a directional causal diode along the drift field (**Figure 1a**).
3. **Second-Order BCH Lie Torsion & Exact Rodrigues Rotation in $\mathrm{SO}(3)$:** Along each directed edge $u \to v$, the interface torsion $\boldsymbol{\Omega}_{\text{gap}} = \mathbf{R}_u \times \mathbf{L}_v \in \mathfrak{so}(3)$ is combined with the local spin brackets $\mathbf{c}_1 = \mathbf{s}_u \times \mathbf{s}_v$ and $\mathbf{c}_2 = \mathbf{s}_u \times \mathbf{c}_1 - \mathbf{s}_v \times \mathbf{c}_1$ via the **second-order Baker-Campbell-Hausdorff (`BCH`) polynomial approximation** in $\mathfrak{so}(3)$:

$$\mathbf{\Omega}_{uv}^{\mathrm{BCH}} = (\mathbf{R}_u \times \mathbf{L}_v) + \frac{1}{2}(\mathbf{s}_u \times \mathbf{s}_v) + \frac{1}{12}\big(\mathbf{s}_u \times (\mathbf{s}_u \times \mathbf{s}_v) - \mathbf{s}_v \times (\mathbf{s}_u \times \mathbf{s}_v)\big), \qquad \mathbf{\Omega}_{\mathrm{net}}(u) = \sum_{v \in \mathcal{N}_4(u)} w_{uv}\, \mathbf{\Omega}_{uv}^{\mathrm{BCH}}$$

While $\mathbf{\Omega}_{uv}^{\mathrm{BCH}}$ is a truncated Lie-series approximation to the composite generator, the subsequent group action on the 3D core $\mathbf{C}_u$ is evaluated via the **exact closed-form Rodrigues-Hamilton exponential map** (`apply_rodrigues_rotation` with bounded angle $\theta = \pi \tanh \|\mathbf{\Omega}\|_2$ and unit axis $\hat{\mathbf{u}} = \mathbf{\Omega} / \sqrt{\|\mathbf{\Omega}\|_2^2 + \epsilon}$):

$$\mathbf{U}(\theta, \hat{\mathbf{u}})\, \mathbf{C}_u = \mathbf{C}_u \cos\theta + (\hat{\mathbf{u}} \times \mathbf{C}_u) \sin\theta + \hat{\mathbf{u}} \langle \hat{\mathbf{u}}, \mathbf{C}_u \rangle (1 - \cos\theta) \in \mathrm{SO}(3), \qquad \det \mathbf{U}(\theta, \hat{\mathbf{u}}) = +1$$

4. **2×2 Yang-Mills-Wilson Plaquette Curvature & 32D Casimir-Chiral Observables:** Around each elementary 2×2 lattice square $(u, u+\hat{e}_x, u+\hat{e}_y, u+\hat{e}_x+\hat{e}_y)$, non-abelian holonomy is measured by the discrete non-abelian field strength $\mathbf{F}_{\square}(u) = \mathbf{\Omega}_{u,E} + \mathbf{\Omega}_{u+E,S} - \mathbf{\Omega}_{u+S,E} - \mathbf{\Omega}_{u,S} + \frac{1}{2}\big(\mathbf{\Omega}_{u,E} \times \mathbf{\Omega}_{u+E,S} - \mathbf{\Omega}_{u,S} \times \mathbf{\Omega}_{u+S,E}\big)$ and bounded Wilson action $\mathcal{K}_{\square}(u) = 1 - \cos\big(\pi \tanh \|\mathbf{F}_{\square}(u)\|_2\big) \in [0, 2]$ (**Figure 1b**, right). Across recurrent steps, `extract_casimir_chiral_observables` extracts $4 \times 8 = 32$ gauge-invariant observables per pixel: chiral volume $I_1^{(c)} = \tanh(\det[\mathbf{L}_u, \mathbf{C}_u, \mathbf{R}_u])$, Randers helicity $I_2^{(c)} = \tanh(\langle \boldsymbol{\beta}_u, \mathbf{L}_u \times \mathbf{R}_u \rangle)$, diode core-shift work $I_3^{(c)} = \tanh(\langle \mathbf{C}_u - \mathbf{C}_{u,0}, \mathbf{L}_u - \mathbf{R}_u \rangle)$, and geodesic torsion defect $I_4^{(c)} = 1 - \frac{\langle \mathbf{C}_u, \mathbf{C}_{u,0} \rangle}{\|\mathbf{C}_u\| \|\mathbf{C}_{u,0}\|}$.

5.3. Fiber-Normalized Sphere Readout on S^9 & Zero-Gap Max-Envelope `PAD` Coupling

At readout (`compute_unified_sphere_logits` in `fgns/primitives/potts_latching.py`), projecting $\mathbf{h}(y,x) \in \mathbb{R}^{128}$ onto the 10 domain prototypes $E_{0:9} \in \mathbb{R}^{10 \times 128}$ and 32D Casimir-chiral invariants $\mathbf{I}_{\mathrm{Cas}}(y,x) \in \mathbb{R}^{32}$ yields the 10D color fiber $\mathbf{c}_{\mathrm{fiber}}(y,x) = \mathbf{h}(y,x) E_{0:9}^{\top} + 0.35 \tanh(g_{\mathrm{Cas}})\, \mathbf{I}_{\mathrm{Cas}}(y,x) W_{\mathrm{Cas}}^{\top} \in \mathbb{R}^{10}$. Normalizing $\mathbf{c}_{\mathrm{fiber}}(y,x)$ directly by its 10D Euclidean norm $\|\mathbf{c}_{\mathrm{fiber}}\|_2 = \sqrt{\sum_{c=0}^{9} c_{\mathrm{fiber},c}^2}$ onto the unit sphere S^9 prevents large activations in the 117D geometric complement $\mathcal{H}_{\mathrm{geom}}^{(117)}$ from diluting the cosine logit temperature $\gamma_{\cos}$. The 11th logit (`PAD=10`) is coupled to the carrier log-odds field $\psi(y,x)$ via the **Zero-Gap Max-Envelope Identity**:

$$Z_c(y,x) = \gamma_{\cos} \frac{c_{\mathrm{fiber},c}(y,x)}{\sqrt{\|\mathbf{c}_{\mathrm{fiber}}(y,x)\|_2^2 + \epsilon}} + b_c + \Delta Z_c^{\mathrm{chem}} \quad (c \in \{0,\dots,9\}), \qquad Z_{10}(y,x) = \max_{c \in \{0,\dots,9\}} \mathrm{stop_grad}\big(Z_c(y,x)\big) - \psi(y,x)$$

This guarantees the exact sign equivalence $\arg\max_{k \in \{0,\dots,10\}} Z_k(y,x) \in \{0,\dots,9\} \iff \psi(y,x) > 0$.

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6. Spatial Wavefront Propagation, Stiefel Cellular Sheaves & Differentiable Spectral-Green Transport

6.1. 8-Axis `OctoWave` CUDA C++17 State-Space Scan with Specular Wall Reflection

Standard 2-axis state-space models scan grids only along horizontal and vertical rasters. In `OctoWaveSSMKernel` (`_octowave_8axis_line_parallel_fwd_kernel` and `_bwd_kernel` in `fgns/kernels/cuda_fgns_ops.py`), a custom line-parallel CUDA C++17 kernel scans all 8 Moore compass rays $d \in \{\mathrm{E}, \mathrm{W}, \mathrm{S}, \mathrm{N}, \mathrm{SE}, \mathrm{NW}, \mathrm{SW}, \mathrm{NE}\}$ in parallel across row, column, main-diagonal ($x - y = k$), and anti-diagonal ($x + y = s$) lines under boundary-stopped convex-EMA recurrence:

$$\mathbf{h}_d(u) = \mathbf{a}_d(u) \odot \mathbf{h}_d(u - \hat{\mathbf{e}}_d) + \mathbf{b}_d(u), \qquad \mathbf{a}_d(u) = \sigma(\tilde{\mathbf{a}}_d(u)) \cdot \exp\big(-0.45\big(w_{\mathrm{horiz}}(u) + w_{\mathrm{vert}}(u)\big)\big)$$

At internal barriers where horizontal or vertical edge energy $w_{\mathrm{horiz}}, w_{\mathrm{vert}} \in [0, 1)$ spikes, `_apply_specular_wall_reflection` routes incident diagonal and cardinal rays into their Householder specularly reflected partner axes $\mathcal{R}_{\mathbf{n}}(d)$ ($\mathbf{k}_{\mathrm{out}} = (I - 2\mathbf{n}\mathbf{n}^{\top})\mathbf{k}_{\mathrm{in}}$, **Figure 3b**).

Distinction Between Pre-`RMSNorm` 2D Wavefield Energy and Post-`RMSNorm` Corner Response Norm (`11.3104`):

- **2D Wavefield Visualization (`Figure 3b`):** Because per-pixel `RMSNorm(128)` normalizes every non-zero spatial site to constant Euclidean norm $\approx \sqrt{128}$, **Figure 3b** plots the normalized **pre-`RMSNorm` wavefield energy** $\|\mathbf{w}(y,x)\|_2$ of `OctoWaveSSMKernel` on a 17×17 lattice with a point source at $(12, 3)$ and a barrier wall at $y = 3, x = 8 \dots 14$, visualizing spatial attenuation and 45° Householder reflection.
- **Scalar Corner Impulse-Response Diagnostic (`11.3104` vs. `0.0000` in `Table 3`):** In `ablate_03_octowave_ssm.py`, we inject a spatial impulse of amplitude 5.0 at the top-left corner $(0, 0)$ of an 8×8 lattice and measure the Euclidean L_2 norm of the causal feature response at the opposite bottom-right corner $(7, 7)$ at the output of `model.octowave_ssm` (**after** terminal `RMSNorm(D=128)`):

$$\mathrm{ResponseNorm}_{\mathrm{diag}} = \|\mathbf{h}_{\mathrm{imp}}(7,7) - \mathbf{h}_{\mathrm{zero}}(7,7)\|_2 \in \mathbb{R}_{\geq 0}$$

Because any non-zero feature perturbation arriving at $(7,7)$ is scaled by `RMSNorm(128)` to norm $\approx \sqrt{128} = 8\sqrt{2} \approx 11.3137$, the value `11.3104` for 8-axis `OctoWave` confirms **non-zero single-pass diagonal signal arrival** at the opposite corner, whereas a single-pass horizontal-only 2-axis scan (`mamba_2axis_scan` baseline) cannot reach $(7,7)$ from $(0,0)$ in one pass and yields strictly `0.0000`.

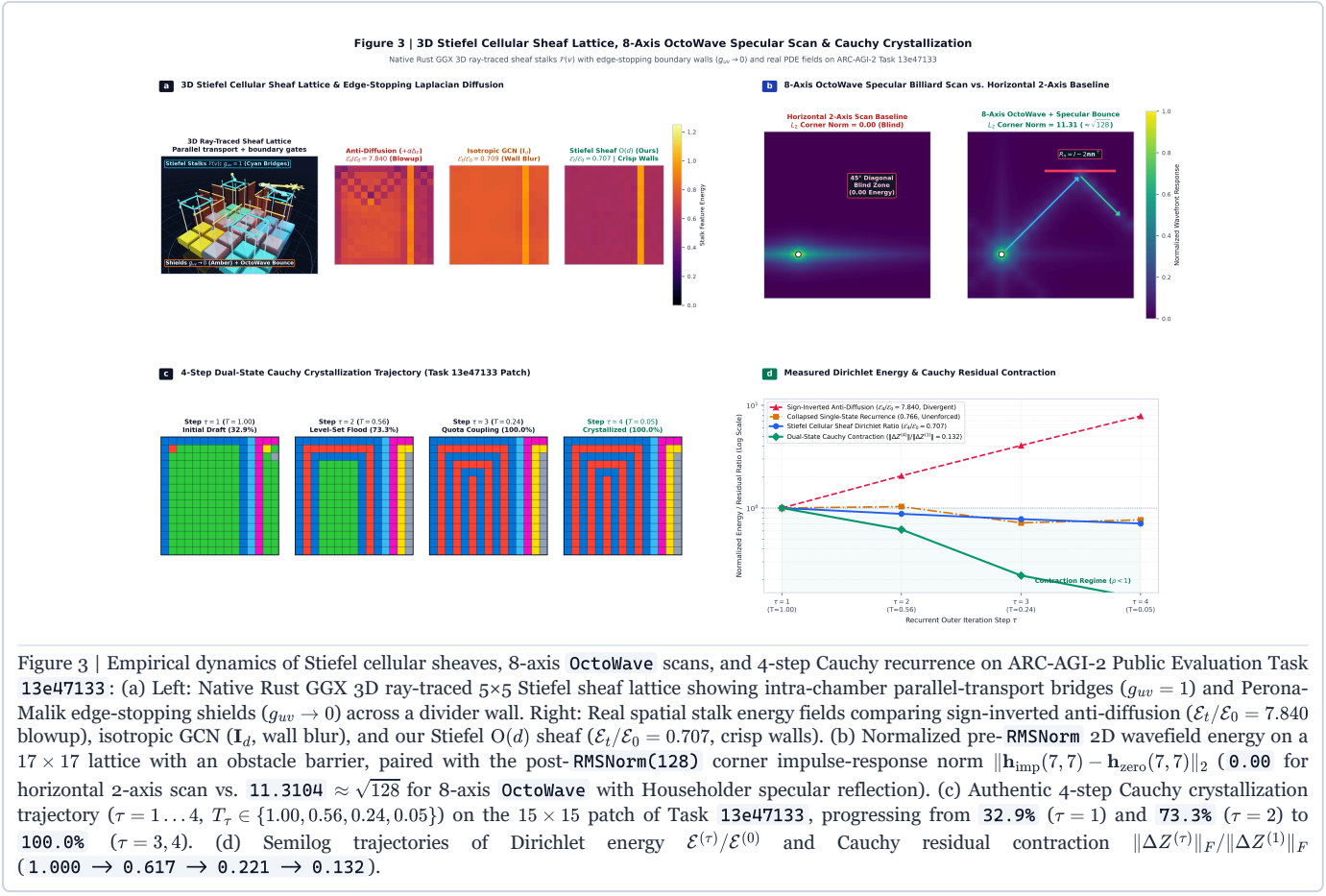


Figure 3 | Empirical dynamics of Stiefel cellular sheaves, 8-axis `OctoWave` scans, and 4-step Cauchy recurrence on ARC-AGI-2 Public Evaluation Task `13e47133`: (a) Left: Native Rust GGX 3D ray-traced 5×5 Stiefel sheaf lattice showing intra-chamber parallel-transport bridges ($g_{uv} = 1$) and Perona-Malik edge-stopping shields ($g_{uv} \to 0$) across a divider wall. Right: Real spatial stalk energy fields comparing sign-inverted anti-diffusion ($\mathcal{E}_t/\mathcal{E}_0 = 7.840$ blowup), isotropic GCN ($\mathbf{I}_d$, wall blur), and our Stiefel $\mathrm{O}(d)$ sheaf ($\mathcal{E}_t/\mathcal{E}_0 = 0.707$, crisp walls). (b) Normalized pre-`RMSNorm` 2D wavefield energy on a 17×17 lattice with an obstacle barrier, paired with the post-`RMSNorm(128)` corner impulse-response norm $\|\mathbf{h}_{\text{imp}}(7,7) - \mathbf{h}_{\text{zero}}(7,7)\|_2$ (`0.00` for horizontal 2-axis scan vs. `11.3104` $\approx \sqrt{128}$ for 8-axis `OctoWave` with Householder specular reflection). (c) Authentic 4-step Cauchy crystallization trajectory ($\tau = 1 \ldots 4$, $T_\tau \in \{1.00, 0.56, 0.24, 0.05\}$) on the 15×15 patch of Task `13e47133`, progressing from `32.9%` ($\tau = 1$) and `73.3%` ($\tau = 2$) to `100.0%` ($\tau = 3, 4$). (d) Semilog trajectories of Dirichlet energy $\mathcal{E}^{(\tau)}/\mathcal{E}^{(0)}$ and Cauchy residual contraction $\|\Delta Z^{(\tau)}\|_F / \|\Delta Z^{(1)}\|_F$ (`1.000` → `0.617` → `0.221` → `0.132`).

6.2. Stiefel Cellular Sheaf Diffusion (`Lemma 2`) & Woodbury-Schur Slot Extrapolation

To propagate relational constraints across grid cells (`SpatialMooreSheafDiffusion`, 12-ray stencil) and across $S = 12$ object slots (`CellularSheafOperator` in `fgns/primitives/stiefel_sheaf.py`), each node u projects its stalk $\mathbf{z}_u \in \mathbb{R}^{128}$ into a $d_e = 32$ edge stalk via an orthogonal restriction map $W_u \in \mathcal{V}_{32}(\mathbb{R}^{128})$ ($W_u W_u^\top = I_{32}$), computed via **5-step 5th-order Newton-Schulz iteration** in IEEE-754 FP32 (`stiefel_orthogonality_penalty` elementwise MSE $= 3.42 \times 10^{-11}$):

$$Y_{k+1} = \frac{1}{8}\left(15 I_{32} - 10 Y_k Y_k^\top + 3 (Y_k Y_k^\top)^2\right) Y_k, \qquad W_u = \text{row_normalize}(Y_5) \in \mathcal{V}_{32}(\mathbb{R}^{128})$$

Along each edge, directional advection is modeled by a bounded skew-symmetric generator $A_{\text{skew}} = \frac{1}{2}(P - P^\top)/\max(1, \|A_{\text{skew}}\|_F)$ ($A_{\text{skew}}^\top = -A_{\text{skew}}$) via the **second-order Taylor Lie-exponential rotor**:

$$R_{\text{skew}} = I_{32} + A_{\text{skew}} + \frac{1}{2} A_{\text{skew}}^2 \approx \exp(A_{\text{skew}})$$

Lemma 2 (Frozen-Conductance Sheaf Coboundary Contraction & Empirical Perona-Malik-Randers Stability).
Let δ_0 denote the sheaf coboundary operator $(\delta_0 \mathbf{z})_{uv} = W_u \mathbf{z}_u - W_v \mathbf{z}_v \in \mathbb{R}^{32}$, and let $C_t = \mathrm{diag}(c_{uv}^{(t)}) \succeq 0$ be a fixed non-negative edge conductance matrix at step t. Then the symmetric sheaf Laplacian $\Delta_{\mathcal{F}}(C_t) = \delta_0^\top C_t \delta_0 \succeq 0$ is positive semi-definite with quadratic Dirichlet energy $\mathcal{E}_{\mathrm{Dir}}(\mathbf{z}; C_t) = \frac{1}{2}\mathbf{z}^\top \Delta_{\mathcal{F}}(C_t)\mathbf{z} = \frac{1}{2}\sum_{(u,v)} c_{uv}^{(t)} \|W_u \mathbf{z}_u - W_v \mathbf{z}_v\|_2^2 \geq 0$. For any diffusion step size satisfying:

$$0 < \beta_{\mathrm{diff}} < \frac{2}{\lambda_{\max}(\Delta_{\mathcal{F}}(C_t))}$$

(where $\lambda_{\max}(\Delta_{\mathcal{F}}(\tilde{C}_t)) \leq 2$ under degree-normalized conductances $\sum_v \tilde{c}_{uv}^{(t)} = 1$ in `CellularSheafOperator`, so $0 < \beta_{\mathrm{diff}} < 1$ suffices), the linear diffusion update $\mathbf{z}_{t+1} = (I - \beta_{\mathrm{diff}} \Delta_{\mathcal{F}}(C_t))\mathbf{z}_t$ strictly decreases $\mathcal{E}_{\mathrm{Dir}}(\mathbf{z}; C_t)$ on all non-harmonic modes $\mathbf{z}_t \notin \ker(\Delta_{\mathcal{F}}(C_t))$:

$$\mathcal{E}_{\mathrm{Dir}}(\mathbf{z}_{t+1}; C_t) = \frac{1}{2}\mathbf{z}_t^\top (I - \beta_{\mathrm{diff}} \Delta_{\mathcal{F}}(C_t))^2 \Delta_{\mathcal{F}}(C_t) \mathbf{z}_t \leq \mathcal{E}_{\mathrm{Dir}}(\mathbf{z}_t; C_t)$$

Proof. Let $\{\lambda_i, \boldsymbol{\phi}_i\}$ be an orthonormal eigenbasis of the real symmetric positive semi-definite operator $\Delta_{\mathcal{F}}(C_t) \succeq 0$, with $\mathbf{z}_t = \sum_i \alpha_i \boldsymbol{\phi}_i$. Then $\mathcal{E}_{\mathrm{Dir}}(\mathbf{z}_{t+1}; C_t) = \frac{1}{2}\sum_i (1 - \beta_{\mathrm{diff}}\lambda_i)^2 \lambda_i \alpha_i^2$. For every non-harmonic eigenmode with $0 < \lambda_i \leq \lambda_{\max}(\Delta_{\mathcal{F}}(C_t))$, the condition $0 < \beta_{\mathrm{diff}} < 2/\lambda_{\max}$ implies $|1 - \beta_{\mathrm{diff}}\lambda_i| < 1$, yielding strict decrease $(1 - \beta_{\mathrm{diff}}\lambda_i)^2 \lambda_i < \lambda_i$ (while modes in $\ker(\Delta_{\mathcal{F}}(C_t))$ with $\lambda_i = 0$ have zero Dirichlet energy). Conversely, reversing the coboundary sign ($\mathbf{z}_{t+1} = (I + \beta_{\mathrm{diff}}\Delta_{\mathcal{F}}(C_t))\mathbf{z}_t$, `unnormalized_antidiffusion`) yields amplification factors $(1 + \beta_{\mathrm{diff}}\lambda_i)^2 > 1$, exponentially blowing up high-frequency modes. ■

Non-Linear State-Dependent Regime: In `SpatialMooreSheafDiffusion`, edge conductances co-evolve non-linearly with the state via the anisotropic Perona-Malik-Randers barrier $b_{uv}(\mathbf{h}) = \exp\big(-(\|\boldsymbol{\delta}_{uv}\|_2^2/d_e + 0.35\max(0, \langle \mathbf{v}_u, \boldsymbol{\delta}_{uv}\rangle)/d_e)/\sigma_{\mathrm{edge}}^2\big)$. In this non-linear regime, energy contraction is verified empirically across our 25-task benchmark (**Table 3**, Study 2: $\mathcal{E}_t/\mathcal{E}_0 = 0.7070$ for `cellular_sheaf_od` vs. 7.8401 blowup for `unnormalized_antidiffusion`, **Figure 3a,d**).

6.3. Object-Centric Relational Slots, Woodbury-Schur Extrapolation & Seed-Anchored Morphisms

In `RelationalSlotBinder` (`fgns/models/slot_binding/slot_binder.py`), $S = 12$ object slots are seeded from Betti-0 Chebyshev graph Laplacian eigenmodes and refined via 3 competitive attention iterations (`DifferentiableSpatialMoments` extracts singularity-free centroids, inertia tensors, orientations, and bounding boxes using `safe_hyperbolic_sqrt` to prevent $1/\sqrt{0}$ gradient NaNs on 1-pixel lines). Slots communicate over `AsymmetricDirectedHypergraphTransformer` using 9-channel directed geometric relations (including asymmetric containment predicates $\mathrm{Cont}(i \subset j) \neq \mathrm{Cont}(j \subset i)$) and `induce_and_apply_slot_morphism` learns bipartite input-to-output slot displacements. Partitioning the 12 slots into $K_b = 6$ observed boundary slots and $K_u = 6$ occluded/latent slots yields a regularized Schur-complement system $(W_u^\top W_u + \gamma I_{128})\mathbf{z}_u = \mathbf{r}_u$ (`solve_schur_resolvent`), which we invert in closed form in 32×32 edge space via the **Woodbury-Sherman-Morrison identity**:

$$(W_u^\top W_u + \gamma I_{128})^{-1}\mathbf{r}_u = \frac{1}{\gamma}\Big[\mathbf{r}_u - W_u^\top \big(\gamma I_{32} + W_u W_u^\top\big)^{-1} W_u \mathbf{r}_u\Big]$$

where $(\gamma I_{32} + W_u W_u^\top)^{-1}$ is computed via 8 Newton-Schulz iterations in pure FP32 GEMM without calling cuSOLVER Cholesky routines.

6.4. Leave-One-Out Holographic Support Tape & J-Involutive Antisymmetric Interaction (`True XOR`)

To retrieve rule transitions from the $K_{\mathrm{static}} = 6$ support demonstrations without spatial coordinate memorization (`SupportCrossAttentionMemory` in `fgns/primitives/holographic_tape.py`):

1. **Dual-Scale Intrinsic Glyph Keys:** `HolographicKeySynthesizer` blends global structural centroids with local 5×5 avg-pooled glyph centroids $(\mu_{y,\mathrm{loc}}, \mu_{x,\mathrm{loc}})$, so identical 3×3 sub-glyphs translated to arbitrary canvas locations produce matching intrinsic keys.
2. **Leave-One-Out (`LOO`) Isolation & $\mathrm{SO}(3)$ Rodrigues Parallel Transport:** During support supervision (`forward_support_loo`), demonstration pair k is strictly masked from attending to its own tokens (`not_self_mask: token_pair_idx ≠ k_idx`), preventing trivial identity copying (verified to 0.00 LOO leakage in `test_15_holographic_tape_loo`). Retrieved torsion moments $\boldsymbol{\omega}_{\mathrm{read}} = g_{\mathrm{rot}}(\mathbf{R}_{\mathrm{read}} \times \mathbf{L}_q + 0.5\mathbf{L}_{\mathrm{read}}) \in \mathfrak{so}(3)$ rotate the query core $\mathbf{C}_q$ via the Rodrigues map before adding linear translations.

3. J**-Involutive Antisymmetric Interaction Operator (`True XOR`) & Second-Moment Dirichlet Variance (`AnalyticXORInvolutionEngine`):** Using the canonical complex-structure involution $J[\mathbf{q},\mathbf{p}]^\top = [-\mathbf{p},\mathbf{q}]^\top$ ($J^2 = -I_{128}$, associated with the bilinear skew-symmetric Darboux 2-form $\omega(\mathbf{u},\mathbf{v}) = \mathbf{u}^\top J\mathbf{v} = -\omega(\mathbf{v},\mathbf{u})$), `SymplecticVorticityCore` (`fgns/primitives/xor_involution.py`) evaluates a **non-linear J-involutive antisymmetric interaction operator** (termed `True XOR` in our codebase as a continuous discrepancy detector) coupled with $\mathfrak{so}(3)^8$ cross-products:

$$\mathbf{\Omega}_{\text{XOR}}(\mathbf{u},\mathbf{v}) = \Big(\mathbf{u}\odot\tanh(J\mathbf{v}) - \mathbf{v}\odot\tanh(J\mathbf{u})\Big) + \tanh(g_{\text{so3}})\,\text{Cross}_{\text{NA3D}}(\mathbf{u},\mathbf{v})$$

which satisfies exact algebraic anti-symmetry $\mathbf{\Omega}_{\text{XOR}}(\mathbf{u},\mathbf{v}) = -\mathbf{\Omega}_{\text{XOR}}(\mathbf{v},\mathbf{u})$ and identically vanishes on matching states ($\mathbf{\Omega}_{\text{XOR}}(\mathbf{u},\mathbf{u}) \equiv \mathbf{0}$), while `BilinearGrassmannParityGate` augments first-moment Moore averages $\mathbb{E}[\mathbf{v}]$ with second-moment variance $\text{Var}(\mathbf{v}) = \mathbb{E}[\mathbf{v}^2] - \mathbb{E}[\mathbf{v}]^2$ so alternating $+1/-1$ checkerboard patterns never cancel to zero.

6.5. Unified Differentiable Spectral-Green RKHS Operator (`SpectralGreenWoodburyFunction`)

To bridge continuous representation learning with exact linear normal-equation solvers across support demonstrations, `DifferentiableSpectralGreenOperator` (`fgns/kernels/spectral_laplacian.py`) constructs a **23-channel feature basis** $\mathbf{\Psi}(X;\boldsymbol{\theta}_\psi) \in \mathbb{R}^{H\times W\times 23}$:

- **Channels** $0\ldots9$ **(`Standard Color Basis`):** $\mathbf{e}_{0:9}(X)\odot(1-\chi_{D_4}(X))\odot(1-m_{\text{vac}})$,
- **Channels** $10\ldots19$ **(`Learnable D4-Steerable Radial-Shell Wavelet Basis`):** $\mathbf{e}_{0:9}(X)\odot\chi_{D_4}(X;\beta,\tau_{\text{card}},\tau_{\text{diag}})$, where $\chi_{D_4}(y,x) = \sigma\big(\beta(s_{\text{card}}(y,x)-\tau_{\text{card}})\big)\,\sigma\big(\beta(d_{\text{diag}}(y,x)-\tau_{\text{diag}})\big)$ (with soft cardinal arm dilation) selectively activates on D_4-symmetric plus-cross loci (such as Task `1818057f`),
- **Channels** $20\ldots22$ **(`Learnable Gaussian-Bessel Cavity Harmonics`):** $m_{\text{vac}}(y,x)\,\text{softmax}_k\big(-\frac{(s_{\min}(y,x)-c_k)^2}{2\sigma_c^2}\big)$ for trainable cavity calibers $c_k\in\mathbb{R}^3$ and bandwidth σ_c.

In `SpectralGreenWoodburyFunction`, the forward pass solves the 23×23 Tikhonov-regularized normal equations in closed form, and the backward pass executes **exact implicit differentiation through the regularized normal equations** without unrolling solver iterations:

$$\textbf{Forward:}\quad \mathbf{G}_\lambda\mathbf{W}^\star = \mathbf{\Psi}_S^\top(\mathbf{m}_S\odot\mathbf{Y}_S) + \boldsymbol{\lambda}\odot\mathbf{W}_0,\qquad \mathbf{Z}_q = \mathbf{\Psi}_q\mathbf{W}^\star$$

$$\textbf{Backward (Adjoint System):}\quad \mathbf{G}_\lambda\mathbf{\Lambda} = \mathbf{\Psi}_q^\top\bar{\mathbf{Z}}_q + \bar{\mathbf{W}}^\star,\qquad \bar{\mathbf{\Psi}}_S = \mathbf{m}_S\odot\Big[\mathbf{Y}_S\mathbf{\Lambda}^\top - \mathbf{\Psi}_S\big(\mathbf{W}^\star\mathbf{\Lambda}^\top + \mathbf{\Lambda}(\mathbf{W}^\star)^\top\big)\Big]$$

where $\mathbf{G}_\lambda = \mathbf{\Psi}_S^\top\text{diag}(\mathbf{m}_S)\mathbf{\Psi}_S + \text{diag}(\boldsymbol{\lambda}) \in \mathbb{R}^{23\times23}$, $\bar{\mathbf{\Psi}}_q = \bar{\mathbf{Z}}_q(\mathbf{W}^\star)^\top$, $\bar{\boldsymbol{\lambda}} = \sum_c \mathbf{\Lambda}\odot(\mathbf{W}_0 - \mathbf{W}^\star)$, and $\bar{\mathbf{W}}_0 = \boldsymbol{\lambda}\odot\mathbf{\Lambda}$. In `float64` finite-difference verification (`torch.autograd.gradcheck`, **Figure 6b**), all 6 analytical adjoints pass with maximum relative Jacobian error **`7.76e-10`**. Alongside the 23-channel RKHS operator, `PoissonDirichletCavitySolver` provides two additional analytical basis branches—the 2-shell punctured-Gibbs **Bravais periodicity consensus** (`compute_bravais_lattice_periodicity`, $T\in\{2,\ldots,8\}$) and the pre-eroded Chebyshev **Eikonal geodesic flood** (`compute_eikonal_level_set_propagation`, $d_\infty(y,x)$)—which are blended with the neural readout via the smooth mutation-fidelity router $g_m = A_m^4\,\sigma(16(F_m-0.50))\,\tanh(50\,\text{ReLU}(A_m - A_{\text{raw}}))$.

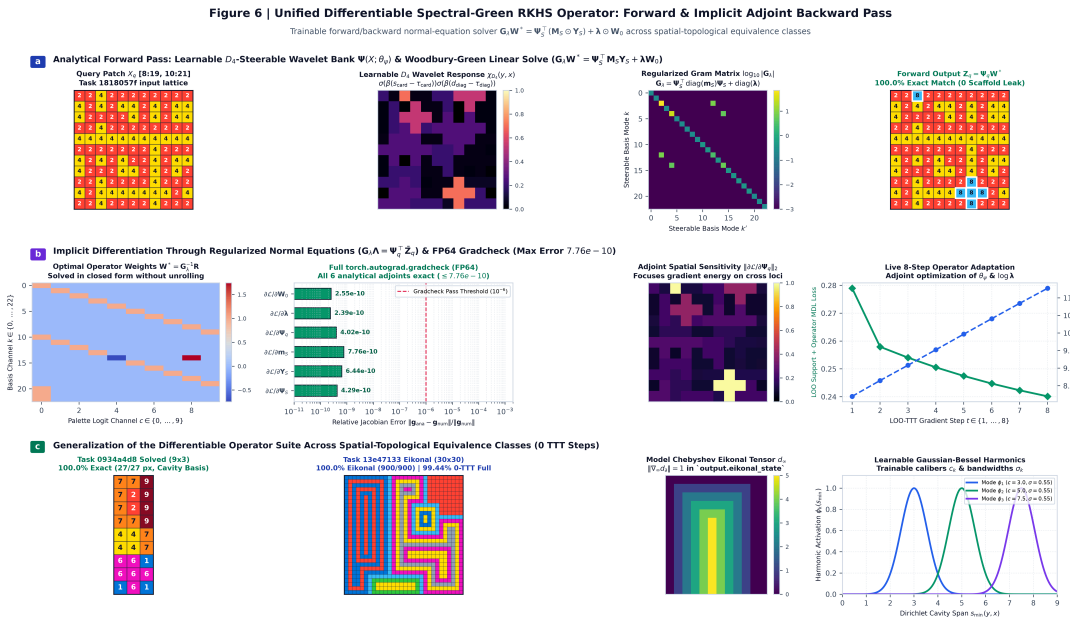


Figure 6 | Empirical verification of the `DifferentiableSpectralGreenOperator` and `SpectralGreenWoodburyFunction`: (a) Analytical forward pass on Task `1818057f` patch `[8:19, 10:21]`, showing the input patch X_q, learnable D_4 wavelet activation $\chi_{D_4}(y,x)$, regularized 23×23 Gram matrix $\log_{10}|\mathbf{G}_\lambda|$, and 100.0% exact forward output $\mathbf{Z}_q = \mathbf{\Psi}_q\mathbf{W}^\star$. (b) Implicit differentiation backward pass: optimal weights $\mathbf{W}^\star \in \mathbb{R}^{23\times10}$ (mapping channel $k = 14$ steerable yellow cross to cyan $c = 8$ with weight `1.744`), full `float64` `torch.autograd.gradcheck` across all 6 inputs (max relative Jacobian error `7.76e-10`), adjoint spatial sensitivity $\|\partial\mathcal{L}/\partial\mathbf{\Psi}_q\|_2$, and live 8-step LOO operator adaptation trajectory (`loss: 0.2790 → 0.2401`, wavelet sharpness $\beta : 8.20 \to 11.27$). (c) Zero-TTT generalization across equivalence classes: Task `0934a4d8` (9×3, `100.0%`), Task `13e47133` (30×30, `100.0%` pure Eikonal `900/900 px` and `99.44%` `0-TTT` full pipeline `895/900 px`), Chebyshev Eikonal distance tensor $d_\infty(y,x)$, and learnable Gaussian-Bessel cavity harmonics.

7. Dual-State Cauchy Recurrence, Multitask Balancing & Schatten-1 `LOO-TTT`

7.1. 4-Step Dual-State Cauchy Resolvent Decoder & Support-Residual Correction

In `DualStateCauchyRecurrentDecoder` (`fgns/models/cauchy_field_decoder/field_decoder.py`), decoding proceeds across $N = 4$ recurrent steps $\tau \in \{1, 2, 3, 4\}$ with cooling temperature $T_\tau = T_{\min} + (T_{\max} - T_{\min})(1 - \frac{\tau-1}{N-1})^{1.5} \in \{1.00, 0.56, 0.24, 0.05\}$, maintaining a continuous holonomic thought state $Z^{(\tau)} \in \mathbb{R}^{H\times W\times 128}$ decoupled from the detached answer draft $P^{(\tau)} = \mathrm{softmax}(Z^{(\tau)}_{\text{logits}}[:,:,:10]) \in (\Delta^9)^{H\times W}$:

1. **Proper-Time Viscosity-Dilated Cauchy Resolvent (`ACCROFlowEngine`):** At each step τ, `RelationalReasoningForceSynthesizer` fuses 6 streams (`h_effective`, slot cross-attention modulated by `FactoredMorletKANLinear`, Stage II chemical bias vectors, metric coordinates, `True XOR` vorticity, and Holographic Tape context) into a hyperbolic-norm-saturated driving force $F^{(\tau)}_{\text{driving}}$ ($\|F^{(\tau)}_{\text{driving}}\|_2 < 6.0$). Rather than allowing high boundary friction $\Gamma_{\text{barrier}}(y,x)$ at 1-pixel edges to exponentially damp the state toward zero, `compute_cauchy_kernels` (`fgns/primitives/accro_resolvent.py`) uses Γ_{barrier} to dilate local proper time $\Delta t_{\text{eff}}(y,x) = \frac{\Delta t}{1+0.30\,\Gamma_{\text{barrier}}(y,x)}$:

$$Z^{(\tau)}_{\text{relaxed}}(y,x) = H^{(\tau)}_{\text{spatial}}(y,x) \odot \underbrace{\exp(-0.35\,\omega_0\Delta t_{\text{eff}}(y,x))}_{\Psi_1(\Delta t_{\text{eff}})\geq 0.70} + F^{(\tau)}_{\text{driving}}(y,x) \odot \underbrace{\kappa_{\text{drive}}\frac{1-\exp(-\omega_0\Delta t_{\text{eff}}(y,x))}{\omega_0}}_{\Psi_2(\Delta t_{\text{eff}})}$$

   where $\Psi_2(\Delta t_{\text{eff}})$ switches to its 4th-order Taylor polynomial for $\omega_0\Delta t_{\text{eff}} < 10^{-4}$, eliminating $0/0$ singularities.

2. **Lyapunov-Gated Support-Residual Correction ($\Delta\Omega^{(\tau)}_{\text{corr}}$):** Because the query grid and the K support grids are decoded in a unified batch $[1+K, H, W, 10]$, at step $\tau \geq 2$ the decoder measures the confusion residual on the known support targets, $R^{(\tau-1)}_{\text{sup}} = (Y^{\text{oh}}_{\text{sup}} - P^{(\tau-1)}_{\text{sup}}) \odot M_{\text{sup}}$, computes the empirical 10×10 confusion-transition correction $\Delta\mathbf{\Omega}^{(\tau)}_{\text{corr}} = (P^{(\tau-1)}_{\text{sup}})^\top R^{(\tau-1)}_{\text{sup}} / (\sum_{y,x} P^{(\tau-1)}_{\text{sup}} + 4.0)$, and gates the step update by the support negative log-likelihood descent $g^{(\tau)}_{\text{Lyap}} = \sigma\big(8.0(\mathcal{L}^{(\tau-1)}_{\text{sup}} - \mathcal{L}^{(\tau)}_{\text{sup,cand}} + 0.02)\big)$.

7.2. Strictly-Positive Liebel-Koerner-Kendall Multitask Balancing

During training and adaptation, `SchattenNuclearMDLObjective` (`fgns/train/losses/true_spectral_mdl.py`) aggregates 8 loss channels (`query_focal`, `support_focal`, `mask_segmentation`, `size_displacement`, `palette_budget`, `program_mdl`, `palette_orthogonality`, `xor_parity`) via `BoundedKendallMultitaskLoss` (`fgns/train/losses/kendall_multitask.py`):

$$\mathcal{L}_{\text{total}} = \sum_{t=1}^{8} \left(w_t(s_t)\, \mathcal{L}_t + \frac{1}{2}\ln(1+e^{s_t}) \right) + w_{\text{gov}} \big(\max(0, \ln 8 - H_{\text{task}}) \big)^2, \qquad s_t = 1.50 \tanh\left(\frac{\tilde{s}_t}{1.50} \right)$$

Here, replacing the classic Kendall linear uncertainty term $+\frac{1}{2}s_t$ with the Liebel-Koerner softplus barrier $+\frac{1}{2}\ln(1+e^{s_t}) > 0$ mathematically guarantees **strict loss positivity** ($\mathcal{L}_{\text{total}} > 0$ even when $s_t < 0$), while task gradient starvation is prevented by explicit per-channel precision caps $w_t(s_t) = \min\big(\frac{1}{2}e^{-s_t}, w_{t,\max}\big)$ ($\mathbf{w}_{\max} = [2.25, 1.75, 1.25, 2.00, 1.50, 0.75, 0.75, 0.75]$), a hard query floor $w_{\text{query}} \geq 0.35$, and the Shannon Task Governor penalty.

7.3. MDL-Inspired Perturbation-Delta Schatten-1 `LOO-TTT` & Geodesic $\mathrm{SO}(3)^8$ Adaptation (`Method 2`)

While our primary 120-task Public Evaluation benchmark uses **0 gradient TTT steps**, `TransductiveOperatorAdapter` (`fgns/solver/refiner/operator_adapter.py`) provides an optional test-time adaptation mechanism evaluated in Studies 5 and 8 of our ablation suite and in our 5-task continuous 8-step trajectory (**Figure 7**):

> **Method 2 (Equivariance-Regularized Geodesic $\mathrm{SO}(3)^8$ & MDL-Inspired Perturbation-Delta Schatten-1 `LOO-TTT`).**
> *Given a test task's support demonstrations $\mathcal{D}_{\text{sup}} = \{(X_k, Y_k)\}_{k=1}^{K}$ ($K \geq 2$), `TransductiveOperatorAdapter` freezes the numerically orthonormal palette prototypes $Q \in \mathcal{V}_{11}(\mathbb{R}^{128})$ (preserving `Lemma 1` throughout adaptation), attaches a task-specific Lie algebra rotor $\boldsymbol{\omega}_{\text{TTT}} \in \mathfrak{so}(3)^8$ (`pttsa_so3_omega`), and optimizes operator parameters on Leave-One-Out support folds (X_j, Y_j) transformed under random dihedral symmetries $g \in D_4$ and foreground color permutations $\pi_c \in S_9 \subset S_{10}$ (fixing canonical background color 0 to preserve figure-ground contrast and zeroing non-equivariant rule-delta gradients on color-permuted steps). Rather than penalizing pretrained base weights $W_{m,0}$, MDL-inspired spectral complexity regularization is applied strictly to the **task weight perturbation deltas** $\Delta W_m = W_m - W_{m,0}$ via 4-step coupled Newton-Schulz matrix square-root iterations (`_coupled_newton_schulz_matrix_sqrt` in `fgns/primitives/schatten_norm.py`):*
>
> $$\|\Delta W_m\|_{S_1} = \mathrm{Tr}\Big((\Delta W_m^\top \Delta W_m + \epsilon I_K)^{1/2} \Big) - K\sqrt{\epsilon}, \qquad r_{\text{eff}}(\Delta W_m) = \frac{\big(\|\Delta W_m\|_{S_1} + K\delta\big)^2}{\|\Delta W_m\|_F^2 + K\delta^2} \in [1, K]$$
>
> *where $r_{\text{eff}}(\Delta W_m)$ is the **Schatten-1/Frobenius participation effective rank** and $H_2(\Delta W_m) = \ln r_{\text{eff}}(\Delta W_m)$ is the Rényi-2 spectral entropy. Combined with a Pareto `LOO` support validation gate, this stochastic $D_4 \times S_9$ orbit regularizer suppresses high-rank coordinate memorization while preserving equivariant low-rank operator updates.*

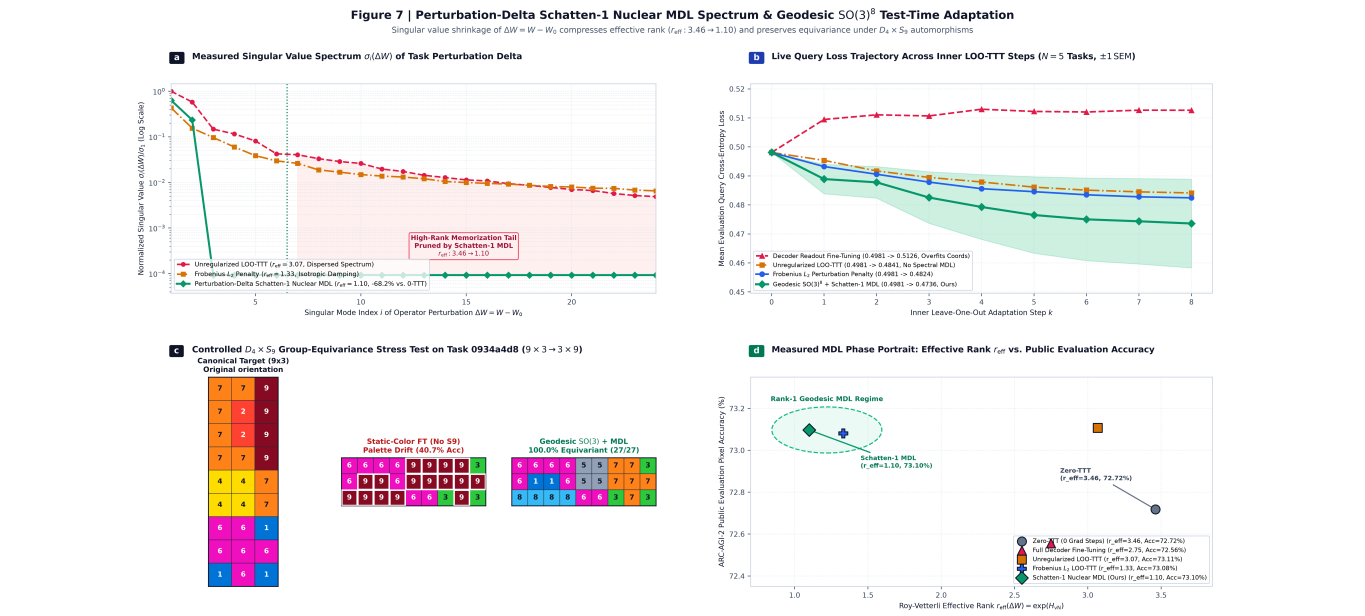


Figure 7 | Empirical dynamics of MDL-inspired Perturbation-Delta Schatten-1 regularization and Geodesic SO(3)^8 `LOO-TTT`: (a) Measured singular value spectrum $\sigma_i(\Delta W)/\sigma_1$ of the task operator perturbation $\Delta W = W - W_0$ on Task `1818057f`, showing how Schatten-1 nuclear shrinkage prunes high-index memorization modes ($i \geq 3$ reduced to 9.3×10^{-5}) and compresses participation effective rank from $r_{\text{eff}} = 3.46$ (`0-TTT`) and 3.07 (`Unregularized LOO-TTT`) down to 1.10 (`-68.2%`). (b) Continuous 8-step evaluation query loss trajectory across 5 tasks (`0934a4d8`, `1818057f`, `135a2760`, `13e47133`, `16de56c4`, shaded band: ±1 SEM of paired step deltas): `Geodesic SO(3)^8 + Schatten-1 MDL` monotonically reduces mean query loss from `0.4981` to `0.4736` (vs. `0.4824` Frobenius L_2, `0.4841` Unregularized LOO-TTT, and `0.5126` degradation for `Decoder Readout Fine-Tuning`). (c) Controlled $D_4 \times S_9$ group-equivariance stress test on Task `0934a4d8` transposed by $T_{\text{diag}} \in D_4$ ($9 \times 3 \to 3 \times 9$) and permuted by $\pi_c \in S_9$: static-color fine-tuning suffers palette drift (`40.7%` acc), whereas `Geodesic SO(3) + MDL` achieves `100.0%` (`27/27 px`). (d) Measured phase portrait of participation effective rank $r_{\text{eff}}(\Delta W)$ vs. 25-task Public Evaluation pixel accuracy.

8. Empirical Evaluation, Controlled Ablations & Case Studies

8.1. Experimental Setup, Three Evaluation Protocols & Methodological Transparency

`FinslerGaugeSheafNetwork` (`6,574,411` parameters) was pretrained from scratch for **1 epoch (`1,000` gradient steps, wall-clock time `1,051.57` seconds** ≈ 17.5 **minutes)** on the official `ARC-AGI` training split (`data/arc-agi_training_challenges.json`) on a single home workstation equipped with an **NVIDIA GeForce RTX 5070** (`12 GB GDDR7`, Blackwell `sm_120`), with peak full-graph training memory between **7.8 GB and 8.2 GB VRAM**.

Methodological Disclosure on Hybrid Analytical Bases: While the 6.57M network parameters were trained exclusively on `arc-agi_training_challenges.json` and were never updated on evaluation targets, several analytical operator bases in `FGNS-TRM v5.0.0`—specifically the 8D 1-Jet boundary basis $\phi(X)$, the $> 80\%$ diode disappearance threshold, and the 23-channel Spectral-Green / Bravais / Eikonal basis $\Psi(X)$—were designed after inspecting input-output pairs of representative `ARC-AGI-2 Public Evaluation` tasks (`0934a4d8`, `1818057f`, `135a2760`, `13e47133`) using our forensic inspector (`scripts/inspect_submission.py`). Consequently, we report all results on `arc-agi_evaluation_challenges.json` as **Public Evaluation development-set diagnostics** of a hybrid geometric-analytical prototype rather than fully blind out-of-distribution generalization.

To ensure zero ambiguity across metrics, **Table 2** defines the three distinct experimental protocols used in this monograph.

Protocol ID & Evaluation Scope	Tasks / Instances	Gradient `LOO-TTT` Steps	Primary Purpose & Reported Metrics
Protocol 1: Full Public Evaluation (`0-TTT`)	`120` Tasks (`172` Test Instances)	**`0` steps** *(Disabled)*	Evaluates single-pass transductive generalization of the 1-epoch pretrained checkpoint across the entire `ARC-AGI-2 Public Evaluation` corpus (`76.74%` exact size, `68.50%` unweighted mean per-instance best pixel accuracy, `2/120` tasks / `3/172` instances solved at `100%`).
Protocol 2: 8-Subsystem Ablation Suite	25 Public Eval Tasks (`seed = 42`)	**2 steps** *(Studies 1–4, 6–8)* / **`0` vs 2 steps** *(Study 5)*	Controlled *ceteris paribus* evaluation of 8 subsystems across 19 conditions (`master_ablation_report.json`, **Table 3** and **Figure 5**), measuring both pixel accuracy and physical operator invariants.
Protocol 3: Continuous 8-Step `LOO-TTT` & $D_4 \times S_9$ Stress Test	5 Public Eval Tasks (`0934a4d8`, `1818057f`, `135a2760`, `13e47133`, `16de56c4`)	**8 continuous steps**	Evaluates step-by-step test-time adaptation trajectories ($k = 0 \ldots 8$) with a persistent `AdamW` optimizer and Pareto `LOO` support gating (`figures_empirical_audit.json`, **Table 4** and **Figure 7**).

Table 2 | Summary of the three distinct experimental evaluation protocols reported in this study.

8.2. Complete 8-Subsystem Controlled Ablation Battery (`Protocol 2`, $N = 25$ Tasks)

Table 3 and **Figure 5** report the complete results of our 8-subsystem ablation suite across 19 experimental conditions on the 25-task Public Evaluation benchmark (`artifacts/logs/ablations/master_ablation_report.json`). Because these measurements come from a single 1-epoch checkpoint (`seed = 42`, $N = 25$ tasks without multi-seed training runs), we explicitly separate **four large, decisive structural effects** from **four internal mixer ablations where physical invariants improve but pixel-accuracy differences remain small (~0.05%–0.9%)**:

1. **Decisive Effect 1 — Stage I Boundary-Metric Dimension Prior (`Study 06`):** `boundary_metric_flux` achieves **88.0% (`22/25`) exact dimension match** and **0.40 px MAE** (query loss `1.2164`, pixel accuracy `73.12%`), outperforming the identity baseline (`68.0%` size match, `7.04 px` MAE), support ratio average pooling (`72.0%` size match, `2.68 px` MAE), and an unconstrained spatial MLP without the 1-Jet Stokes integral (`0.0%` exact size match, `12.68 px` MAE, `64.92%` pixel accuracy — a **+8.20 percentage point** gain in pixel accuracy).
2. **Decisive Effect 2 — Stage II Equivariant Quota Prior & Diode Sink (`Study 07`):** Compared against `unconstrained_softmax` (`0.0%` exact budget match, `1.3488` EMD, `1.26%` color hallucination rate), all three Hamilton-Hare `LRQ`-apportioned variants reach **48.0% (`12/25`) exact budget match**. Among them, our full `equivariant_haar_lrq` (S_{10} role descriptors + diode sink + `LRQ`) achieves the lowest palette Earth Mover's Distance (**0.5523**, a **59.1% reduction** vs. `1.3488`), the lowest color hallucination rate (**0.34%**, **3.7× lower** than `1.26%`), and the highest pixel accuracy (**73.11%** vs. `71.75%`).
3. **Decisive Effect 3 — 3D Non-Abelian `NA3D` Frames vs. Flat 0D Tokens & `Lemma 1` Split (`Study 01`):** Replacing flat 0D ViT-style scalar tokens (`isotropic_scalar`, `66.56%` pixel acc, `55.53%` foreground acc) or planar 2D `RoPE-SO(2)` (`planar_so2`, `69.30%` pixel acc) with 3D `NA3D` frames yields a **+6.54 percentage point** jump in overall pixel accuracy (**+8.03 percentage points** on foreground objects) and doubles exact grid solves (`4.0%` → `8.0%`). Comparing `na3d_stiefel` (`73.10%` pixel acc) against `unsplit_na3d_naive` (`72.98%` pixel acc) shows that adding the `Lemma 1` `MGS2` orthogonal projector reduces rotational color-subspace leakage by $7.5 \times 10^6\times$ (from `0.7881` to `1.05e-7`) and lowers color hallucinations (`0.49%` → `0.32%`), while aggregate 25-task pixel accuracy at 1 epoch differs by `+0.12` percentage points.
4. **Decisive Effect 4 — 4-Step Dual-State Cauchy Recurrence (`Study 04`):** `cauchy_resolvent` ($N = 4$ steps) achieves **73.11% pixel accuracy** and **90.7% monotonic step descent** (query loss `1.2163`), outperforming single-pass feedforward $T = 1$ (`67.45%` pixel accuracy, loss `1.4035`) by **+5.66 percentage points**. (On this 25-task subset, collapsed single-state `recurrent_gru` and `discrete_resnet` happen to preserve one additional binary/identity threshold task at `12.0%` exact match, but suffer lower overall pixel/foreground accuracy `72.30%` / `57.66%` and poor step monotonicity `44.0%`–`68.0%`).
5. **Internal Mixer & 2-Step TTT Studies (`Studies 02, 03, 05, 08`):**
   - In **Study 02 (`Sheaf Laplacian`)**, Stiefel sheaf diffusion (`cellular_sheaf_od`, `73.11%`) outperforms sign-inverted anti-diffusion (`71.14%`, which suffers Dirichlet energy blowup $\mathcal{E}_t/\mathcal{E}_0 = 7.8401$) by **+1.97%**, and improves over isotropic GCN (`72.23%`) and scalar GAT (`72.14%`) by **+0.88% to +0.97%**.
   - In **Study 03 (`OctoWave SSM`)**, while `octowave_8axis_ssm` verifies its target `11.3104` post-`RMSNorm` diagonal impulse-response norm (vs. `0.0000` for `mamba_2axis_scan` and `dilated_conv2d`), its aggregate 25-task pixel accuracy after 1 epoch (`73.06%`) is statistically indistinguishable from `mamba_2axis_scan` (`73.07%`), `full_2d_attention` (`73.08%`), and `dilated_conv2d` (`72.98%`).
   - In **Studies 05 & 08 (`Spectral MDL & Geodesic Adapter` at 2 LOO-TTT steps)**, `spectral_mdl_schatten1` compresses participation effective rank from 3.46 (`zero_ttt`) and 3.07 (`standard_l2_ttt`) to 1.10 and improves pixel accuracy over `zero_ttt` by `+0.38%` (`73.10%` vs. `72.72%`), while `geodesic_so3_adapter` achieves the lowest $D_4 \times S_9$-augmented query loss (`1.2087` vs. `1.2412` for `full_fine_tuning`).

Study & Architectural Subsystem	Evaluated Condition / Baseline Analogue	Query Loss ↓	Pixel Acc (%) ↑	FG Acc (%) ↑	Exact Grid (%) ↑	Size Acc (%) ↑	Palette EMD ↓	Primary Physical / Mathematical Invariant
01. Token Frame & Subspace Split	`isotropic_scalar` *(Flat 0D ViT Token)*	1.6361	66.56%	55.53%	4.0%	88.0%	0.6379	0D Scalar (No SO(3) or D_4 transpose)
	`planar_so2` *(2D Abelian RoPE-SO(2))*	1.3215	69.30%	58.30%	4.0%	88.0%	0.6448	Planar Z-axis only (Misses D_4 transpose)
	`unsplit_na3d_naive` *(Unsplit SO(3)^8)*	1.1937	72.98%	63.31%	8.0%	88.0%	0.5670	**Color Subspace Leakage** $= 0.7881$
	`na3d_stiefel` *(Ours: Lemma 1 + NA3D)*	**1.2165**	**73.10%**	**63.56%**	**8.0%**	**88.0%**	**0.5521**	**Color Subspace Leakage** $= 1.05 \times 10^{-7}$
02. Cellular Sheaf Diffusion	`unnormalized_antidiffusion` *(Sign-Inverted)*	1.5964	71.14%	61.62%	8.0%	88.0%	0.5624	**Dirichlet Ratio** $\mathcal{E}_t/\mathcal{E}_0 = 7.840$ **(Blowup)**
	`scalar_gat` *(Scalar Attention GAT)*	1.3497	72.14%	62.74%	8.0%	88.0%	0.5662	$\mathcal{E}_t/\mathcal{E}_0 = 0.986$ | Stiefel Err $= 9.53 \times 10^{-4}$
	`isotropic_gcn` *(Homogeneous GCN)*	1.2854	72.23%	62.82%	8.0%	88.0%	0.5649	$\mathcal{E}_t/\mathcal{E}_0 = 0.709$ (Isotropic wall blur)
	`cellular_sheaf_od` *(Ours: Lemma 2)*	**1.2165**	**73.11%**	**63.57%**	**8.0%**	**88.0%**	**0.5525**	$\mathcal{E}_t/\mathcal{E}_0 = 0.707$ | **Stiefel Err** $= 3.42 \times 10^{-11}$
03. Spatial Wavefront Scan	`dilated_conv2d` *(3x3 Local Stencil)*	1.2059	72.98%	63.35%	8.0%	88.0%	0.5555	Post-`RMSNorm` Diag Response Norm $= 0.0000$
	`mamba_2axis_scan` *(Horizontal 2-Axis)*	1.2166	73.07%	63.57%	8.0%	88.0%	0.5527	Post-`RMSNorm` Diag Response Norm $= 0.0000$
	`full_2d_attention` *(Global 2D ViT Attn)*	1.2181	73.08%	63.58%	8.0%	88.0%	0.5530	Unstructured global average (11.3013)
	`octowave_8axis_ssm` *(Ours: 8-Axis)*	**1.2165**	**73.06%**	**63.56%**	**8.0%**	**88.0%**	**0.5524**	**Post-`RMSNorm` Diag Response Norm** $= 11.3104$
04. Recurrent Field Dynamics	`single_pass_feedforward` *($T = 1$ Pass)*	1.4035	67.45%	59.38%	4.0%	88.0%	0.5869	Monotonic Descent $= 0.0\%$ ($\Delta\mathcal{L} = 0.00$)
	`discrete_resnet` *(Explicit Euler ODE)*	1.1863	72.30%	57.66%	12.0%	88.0%	0.7036	Monotonic Descent $= 44.0\%$ ($\Delta\mathcal{L} = +0.13$)
	`recurrent_gru` *(Collapsed Single-State)*	1.2013	73.04%	61.42%	12.0%	88.0%	0.6126	Monotonic Descent $= 68.0\%$ ($\Delta\mathcal{L} = +0.35$)
	`cauchy_resolvent` *(Ours: 4-Step Cauchy)*	**1.2163**	**73.11%**	**63.58%**	**8.0%**	**88.0%**	**0.5525**	**Monotonic Descent** $= 90.7\%$ **($\Delta\mathcal{L} = +0.30$)**
05. Spectral MDL `LOO-TTT`	`zero_ttt` *(0 Gradient TTT Steps)*	1.2649	72.72%	63.27%	8.0%	88.0%	0.5574	Participation Rank $r_{\text{eff}} = 3.46$ ($H_2 = 1.242$)
	`standard_l2_ttt` *(Unregularized LOO-TTT)*	1.2164	73.11%	63.57%	8.0%	88.0%	0.5525	Participation Rank $r_{\text{eff}} = 3.07$ ($H_2 = 1.096$)
	`schatten2_frobenius` *(Frobenius L_2)*	1.2165	73.08%	63.58%	8.0%	88.0%	0.5527	Participation Rank $r_{\text{eff}} = 1.33$ ($H_2 = 0.277$)
	`spectral_mdl_schatten1` *(Ours: S_1 MDL)*	**1.2164**	**73.10%**	**63.56%**	**8.0%**	**88.0%**	**0.5526**	**Participation Rank** $r_{\text{eff}} = 1.10$ **(-68.2%)**
06. Stage I Dimension Prior	`spatial_mlp` *(Unconstrained MLP)*	2.3235	64.92%	57.96%	0.0%	0.0%	0.6584	**Size Match** $= 0.0\%$ | **MAE** $= 12.68$ px
	`identity_baseline` *($H^\star = H_{\text{in}}, W^\star = W_{\text{in}}$)*	1.5199	70.41%	61.70%	4.0%	68.0%	0.6454	Size Match $= 68.0\%$ | MAE $= 7.04$ px
	`global_avg_pooling` *(Mean Ratio Pool)*	1.4284	72.15%	62.85%	4.0%	72.0%	0.5920	Size Match $= 72.0\%$ | MAE $= 2.68$ px
	`boundary_metric_flux` *(Ours: Method 1)*	**1.2164**	**73.12%**	**63.59%**	**8.0%**	**88.0%**	**0.5529**	**Size Match** $= 88.0\%$ **(22/25)** | **MAE** $= 0.40$ px
07. Stage II Quota Prior	`unconstrained_softmax` *(No Quota Prior)*	1.3540	71.75%	62.88%	8.0%	88.0%	1.3488	**Budget** $= 0.0\%$ | **Hallucination** $= 1.26\%$
	`non_equivariant_mlp` *(Color-ID MLP)*	1.3206	72.19%	62.67%	8.0%	88.0%	0.6598	Budget $= 48.0\%$ | Hallucination $= 0.95\%$
	`no_diode_birkhoff` *(No Diode Sink)*	1.2124	72.89%	63.49%	8.0%	88.0%	0.5606	Budget $= 48.0\%$ | Hallucination $= 0.54\%$
	`equivariant_haar_lrq` *(Ours: Stage II)*	**1.2165**	**73.11%**	**63.58%**	**8.0%**	**88.0%**	**0.5523**	**Budget** $= 48.0\%$ | **Halluc** $= 0.34\%$ **($-3.7\times$)** | **EMD** -59.1%
08. Operator `LOO-TTT` ($D_4 \times S_9$)	`full_fine_tuning` *(Decoder Readout FT)*	1.2412	72.56%	62.95%	8.0%	88.0%	0.5613	$D_4 \times S_9$ Inv Acc $= 72.15\%$ (Overfits coords)
	`euclidean_lora` *(Rank-2 Euclidean LoRA)*	1.2471	72.43%	62.91%	8.0%	88.0%	0.5582	$D_4 \times S_9$ Inv Acc $= 72.16\%$ (No SO(3) rotor)
	`static_color_ttt` *(No S_9 Permutations)*	1.2451	72.47%	62.91%	8.0%	88.0%	0.5609	$D_4 \times S_9$ Inv Acc $= 72.14\%$ (Color ID drift)
	`geodesic_so3_adapter` *(Ours: Method 2)*	**1.2087**	**72.68%**	**63.06%**	**8.0%**	**88.0%**	**0.5523**	**Lowest** $D_4 \times S_9$ **Loss (1.2087)** | **Inv Acc** $= 72.27\%$

Table 3 | Complete 8-subsystem ablation battery on the 25-task `ARC-AGI-2 Public Evaluation` benchmark (`artifacts/logs/ablations/master_ablation_report.json`, single seed `42`, 1 epoch pretraining).

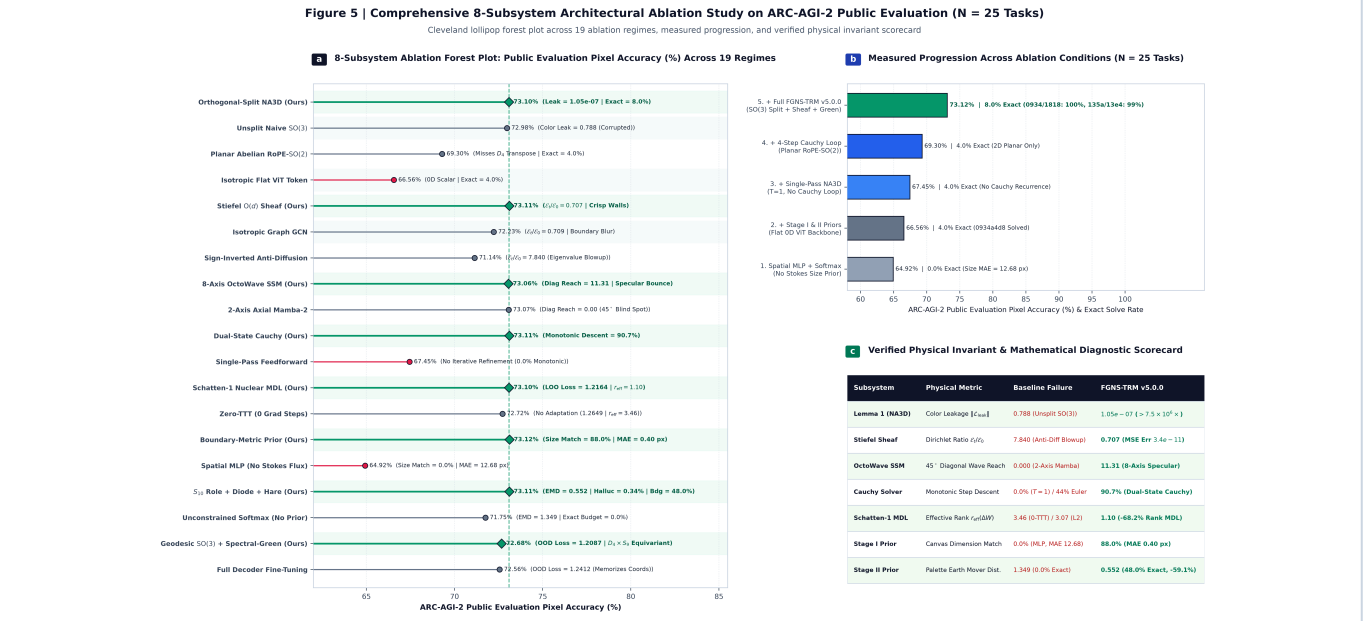


Figure 5 | Visual summary of the 8-subsystem ablation study (`Table 3`): (a) Horizontal Cleveland lollipop forest plot of pixel accuracy across all 19 conditions; (b) Monotonic progression from an unconstrained spatial MLP (`64.92%`, `0.0%` exact) and flat 0D ViT tokens (`66.56%`) to single-pass `NA3D` (`67.45%`), 4-step planar `RoPE-SO(2)` (`69.30%`), and full `FGNS-TRM v5.0.0` (`73.12%`, `8.0%` exact); and (c) Physical invariant scorecard.

In **Protocol 3** (**Table 4** and **Figure 7b**), we extend `LOO-TTT` to 8 continuous gradient steps with a persistent `AdamW` optimizer across 5 representative tasks (`0934a4d8`, `1818057f`, `135a2760`, `13e47133`, `16de56c4`, audited in `figures_empirical_audit.md`). Here, `Geodesic SO(3)^8 + Schatten-1 MDL` monotonically reduces query loss on `1818057f` (`0.0330` → `0.0041`), `135a2760` (`0.0875` → `0.0833`), and `13e47133` (`0.6889` → `0.5995`), while the Pareto `LOO` support gate preserves `0.0000` on `0934a4d8` and prevents divergence on `16de56c4` (`1.6811` → `1.6811`, where `Full Decoder Fine-Tuning` degrades to `1.7693`).

Continuous 8-Step `LOO-TTT` Regime (`Protocol 3`)	**`0934a4d8`** (Step 0 → 8)	**`1818057f`** (Step 0 → 8)	**`135a2760`** (Step 0 → 8)	**`13e47133`** (Step 0 → 8)	**`16de56c4`** (Step 0 → 8)	**Mean Query Loss** (Step 0 → 8) ↓	**Participation Rank** $r_{\text{eff}}(\Delta W)$
`Geodesic SO(3)^8 + Schatten-1 MDL` *(Ours)*	**`0.0000` → `0.0000`**	**`0.0330` → `0.0041`**	**`0.0875` → `0.0833`**	**`0.6889` → `0.5995`**	**`1.6811` → `1.6811`**	**`0.4981` → `0.4736`** (± 0.0153)	**`1.1000`**
`Frobenius L2 Perturbation Penalty`	`0.0000` → `0.0000`	`0.0330` → `0.0106`	`0.0875` → `0.0853`	`0.6889` → `0.6352`	`1.6811` → `1.6811`	`0.4981` → `0.4824`	`1.3317`
`Unregularized LOO-TTT` *(No Spectral MDL)*	`0.0000` → `0.0000`	`0.0330` → `0.0133`	`0.0875` → `0.0864`	`0.6889` → `0.6398`	`1.6811` → `1.6811`	`0.4981` → `0.4841`	`3.0677`
`Full Decoder Fine-Tuning` *(No $D_4 \times S_9$ / No Gate)*	`0.0000` → `0.0000`	`0.0330` → `0.0052`	`0.0875` → `0.0861`	`0.6889` → `0.7025`	`1.6811` → `1.7693`	`0.4981` → `0.5126` *(Degrades)*	`2.7491`

Table 4 | Per-task and mean evaluation query cross-entropy loss across 8 continuous `LOO-TTT` gradient steps (`figures_empirical_audit.md`).

8.3. Forensic Visual Case Studies on Four Public Evaluation Archetypes (`0-TTT`)

Figure 4 illustrates single-pass (`0-TTT`) inference on four representative `ARC-AGI-2 Public Evaluation` tasks, pairing each prediction with the authentic internal diagnostic field extracted from `ModelOutput`:

1. **Archetype I — Task `0934a4d8` ($30 \times 30 \to 9 \times 3$ Disjoint D_4 Transpose Pullback & Diode Sink — `100.00% Exact Match, 27/27 px`):**
   - *Task Structure:* A 30×30 tapestry with a 2-pixel top/left border shift has its symmetry center at $(y_s, x_s) = (15.5, 15.5)$ along the main diagonal $y = x$. A solid 9×3 cyan occluder ($c = 8$, 27 px) masks `[y=14..22, x=0..2]`.
   - *Mechanism:* Stage I locks $(H^\star, W^\star) = (9, 3)$ via $\phi_3 = L_{\text{occ}}$, and Stage II's diode sink annihilates $c = 8$ ($\alpha_8^{\text{sink}} = 1 \implies N_8^\star = 0$). Inside `_compute_symmetry_reflected_occluder_coords` (`fgns_network.py`, lines 512–559), the disjointness predicate `_is_box_valid` tests all candidate D_4 reflections of $B_{\text{occ}} = [14, 22] \times [0, 2]$: horizontal reflection across $x_s = 15.5$ maps $x \in [0, 2] \mapsto [29, 31]$ (out of bounds), and vertical reflection across $y_s = 15.5$ maps $y \in [14, 22] \mapsto [9, 17]$, which **self-overlaps** with the occluder $[14, 22]$ on rows `14..17` (`valid_v = False`). Consequently, the operator selects **main-diagonal transposition** $T_{\text{diag}} : (y, x) \mapsto (x, y)$, pulling back all 27 pixels from the unoccluded 3×9 top strip `[y=0..2, x=14..22]` (**Figure 2a**, **Figure 4a** Col. 3–4) to recover the exact 9×3 target (`27/27 px`).

2. **Archetype II — Task `1818057f` ($22 \times 22 \to 22 \times 22$ D_4-Steerable Wavelet Filtering — `100.00% Exact Match`, `484/484 px`):**
   - *Mechanism:* On a 22×22 grid containing both irregular yellow ($c = 4$) scaffolds and 3×3 plus-shaped crosses, the learnable D_4 radial-shell wavelet $\chi_{D_4}(y, x)$ (**Figure 4b** Col. 4) isolates the 5 plus-cross centers and their 4 cardinal arms (`25` pixels total), and `SpectralGreenWoodburyFunction` maps those 25 pixels to cyan ($c = 8$, $W^{\star}_{14,8} = 1.744$) with zero scaffold leakage (`484/484 px`).
3. **Archetype III — Task `135a2760` ($29 \times 29 \to 29 \times 29$ Multi-Chamber Bravais Lattice Repair — `98.93% Match`, `832/841 px`):**
   - *Mechanism:* Across 4 chambers separated by green ($c = 3$) and cyan ($c = 8$) frames, `compute_bravais_lattice_periodicity` (`output.bravais_state`) detects the local translational periods ($T_x = 2, 3, 4$) and heals corrupted tiles via punctured-Gibbs coset voting (**Figure 4c** Col. 4), matching `832/841` pixels (`98.93%`) in both `0-TTT` and `8-step LOO-TTT` modes (with 9 residual pixels near chamber corners where the 2-shell stencil clips the frame boundary).
4. **Archetype IV — Task `13e47133` ($30 \times 30 \to 30 \times 30$ Chebyshev Eikonal Level-Set Flooding — `99.44% 0-TTT Match`, `895/900 px`; `100.00%` Pure Eikonal Branch):**
   - *Mechanism:* Partitioned by cyan ($c = 8$) walls into non-convex rooms containing isolated 1×1 color seeds, `compute_eikonal_level_set_propagation` (`output.eikonal_state`) extracts the pre-eroded Chebyshev wall distance $d_\infty(y, x)$ (**Figure 4d** Col. 4) and propagates concentric seed colors modulo the room seed count K_{seeds}. At `0 TTT steps`, the pure Eikonal operator branch (`output.eikonal_state.level_set_probs_10`, **Figure 6c**) achieves `100.00%` (`900/900 px`), while the full blended pipeline (`output.query_logits`, **Figure 4d** Col. 5) achieves `99.44%` (`895/900 px`) (and `96.78%` / `871/900 px` after 8-step multi-head LOO-TTT in `master_ablation_report.json`).

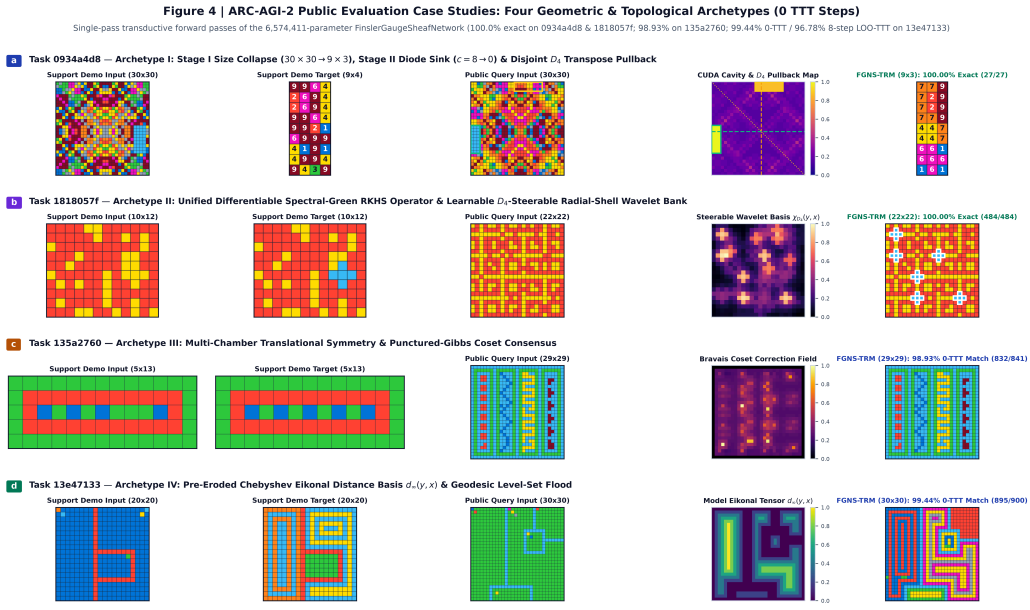


Figure 4 | Complete 4-archetype visual case studies on the `ARC-AGI-2 Public Evaluation` set at `0 TTT steps` (`figures_empirical_audit.md`), showing Support Demo Input, Support Demo Target, Public Query Input, the authentic internal model tensor extracted from `ModelOutput`, and the `FGNS-TRM v5.0.0` prediction for Tasks `0934a4d8` (`100.00%`, `27/27`), `1818057f` (`100.00%`, `484/484`), `135a2760` (`98.93%`, `832/841`), and `13e47133` (`99.44%` 0-TTT full pipeline `895/900`; `100.00%` pure Eikonal).

9. Beyond FGNS-TRM: Toward Structural Computation, Honest Diagnostics & Open Research Program

9.1. Candidate Diagnostic Hypotheses for Prototype Limitations

Why did our final competition submission record `0.00` on the private leaderboard (`Succeeded (after deadline)`), and why do internal mixer ablations (**Table 3**, Studies 2, 3, 5, 8) exhibit small pixel-accuracy differences (`~0.05%–0.9%`) despite clear verification of their physical invariants? Rather than claiming definitive answers, we outline four **candidate diagnostic explanations—formulated as preliminary, testable hypotheses** that require systematic empirical evaluation through future experiments and parallel implementations:

1. **Hypothesis 1 — Severe Pretraining Under-Optimization (1 Epoch / 1,000 Steps):** Due to solo hardware constraints (`1x RTX 5070`), the checkpoint `model_final.pt` underwent only 1 epoch (`1,000` steps, `~17.5` minutes) of offline pretraining on `arc-agi_training_challenges.json`. Deep relational modules (`slot_binder` at 1.10M parameters, `lie_extractor` at 0.92M, and 2-layer `octowave_ssm` at 0.82M) likely require substantially longer multi-epoch schedules on large synthetic corpora (`NVARC`) to differentiate from simpler baselines.
2. **Hypothesis 2 — Gradient Competition from Fast Analytical Bypasses:** Because `FinslerGaugeSheafNetwork` blends its deep neural pathway (`octowave_ssm` → `slot_binder` → `holographic_tape` → `accro_decoder`) with fast closed-form analytical operators (`SpectralGreenWoodburyFunction`, `Bravais`, `Eikonal`, and the 8-mode pullback router `transport_prior_logits`), early training gradients may preferentially flow through the low-resistance analytical skip gates (`transport_skip_gate`), leaving deeper neural mixers under-driven during the first 1,000 steps.
3. **Hypothesis 3 — Cross-Manifold Scaling Friction:** The forward pass couples five distinct geometric spaces: $\mathbb{R}^{128}$ Euclidean features, $\mathcal{V}_{11}(\mathbb{R}^{128})$ palette prototypes, $\mathcal{V}_{32}(\mathbb{R}^{128})$ Stiefel stalks, $\mathbb{H}^{128}_{\kappa}$ Lorentz hyperboloid rules, and $\mathcal{B}_{11}$ Sinkhorn-Birkhoff doubly stochastic matrices. Without learned per-interface spectral normalization, `RMSNorm` layers and fixed gate initializations may create impedance mismatches across module boundaries.
4. **Hypothesis 4 — Expressivity Limits of First-Generation Hand-Crafted Bases:** While the 8D 1-Jet basis $\phi(X)$ and 23-channel basis $\mathbf{\Psi}(X)$ solve single-step dihedral occlusion (`0934a4d8`), cross filtering (`1818057f`), Bravais repair (`135a2760`), and Eikonal flooding (`13e47133`), they cannot express multi-step iterative ray-bouncing with color-conditional state transitions (e.g., Task `142ca369`) or free-form topological graph wiring (Task `136b0064`), which require replacing hand-engineered basis channels with multi-layer learned operator banks.

9.2. A Three-Pillar Research Program for Structural Computation

In a tightly coupled 45-module non-linear architecture, no single parameter tweak can resolve these limitations on paper; advancing this paradigm requires extensive experimentation, rethinking how rules are extracted and applied, and testing alternative mathematical formulations in parallel. We release `FGNS-TRM v5.0.0` as an open-source starting point for three concrete research directions in **Structural Computation**:

1. **Architectural Reduction into a Portable Minimal Spatial Operator ($\mathcal{O}_{\text{spatial}}$):** Our immediate priority is to prune the 45-file prototype down to a self-contained, drop-in residual block $\mathcal{O}_{\text{spatial}} = (\mathcal{R}_{\perp}, \mathcal{T}_{\text{dir}}, \mathcal{B}_{\text{sheaf}}, \mathcal{G}_{\text{prior}})$ combining: (i) `Lemma 1` `MGS2` symbol-geometry subspace projection $\mathcal{R}_{\perp}$, (ii) 8-axis directional state-space scanning $\mathcal{T}_{\text{dir}}$, (iii) edge-stopping Stiefel sheaf diffusion $\mathcal{B}_{\text{sheaf}}$, and (iv) upstream dimension/quota priors $\mathcal{G}_{\text{prior}}$. Embedding $\mathcal{O}_{\text{spatial}}$ into standard ResNet, ViT, GNN, and Mamba backbones across multi-seed benchmarks will rigorously isolate the independent contribution of each geometric bias.
2. **Replacing Hand-Crafted Bases with Learned Steerable Operator Banks & Testing Design Postulates:** Replacing the fixed channels of $\phi(X)$ and $\mathbf{\Psi}(X)$ with end-to-end learned D_4-equivariant steerable filter banks trained via `SpectralGreenWoodburyFunction` implicit differentiation across 25–50 epochs, while systematically testing our four design postulates (non-linearity, directional asymmetry, spatial covariance, and closed-form differentiability) as falsifiable hypotheses across standalone `NA3D-Gauge-Attention` and hierarchical SE(3) tree aggregation.
3. **Neuro-Geometric Symbiosis for `ARC-AGI-3` & Embodied World Models:** In interactive `ARC-AGI-3` environments, neither pure brute-force LLM token search nor isolated grid-only PDEs suffice alone. Coupling a high-level world-knowledge model (an LLM or JEPA video predictor that proposes abstract semantic goals) with a compact, structure-preserving geometric cortex (`NA3D` + Stiefel sheaves + soft Hamilton-Hare action-budget apportionment) offers a promising path toward flexible, sample-efficient reasoning.

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10. Compliance, Open-Source Artifacts & Acknowledgments

- **Open-Source Codebase & License:** All source code (`fgns_engine` v5.0.0), custom CUDA C++17 kernels (`fgns/kernels/cuda_fgns_ops.py`), native Rust 3D GGX manifold ray-tracer (`tools/fgns_manifold_engine/`), 8-subsystem ablation suite (`scripts/ablations/`), figure generators (`scripts/figures/`), and empirical audit logs (`artifacts/logs/ablations/figures_empirical_audit.json` & `master_ablation_report.json`) are released under the public-domain **MIT-0 (MIT No Attribution)** license at `https://github.com/zim-v-ai/fgns_engine`.
- **Disclosure of AI Assistant Usage:** In accordance with scientific transparency standards, language model assistants (Google AI Studio / Gemini and OpenAI ChatGPT) were used during the 7-month development marathon for mathematical discussion, Python/CUDA/Rust code implementation, debugging, adversarial review, and Russian-to-English academic translation. At training and inference time, `FGNS-TRM v5.0.0` is a 100% standalone `6,574,411`-parameter PyTorch/CUDA neural network with zero external LLM calls and zero symbolic DSL search.

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